

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

Complete

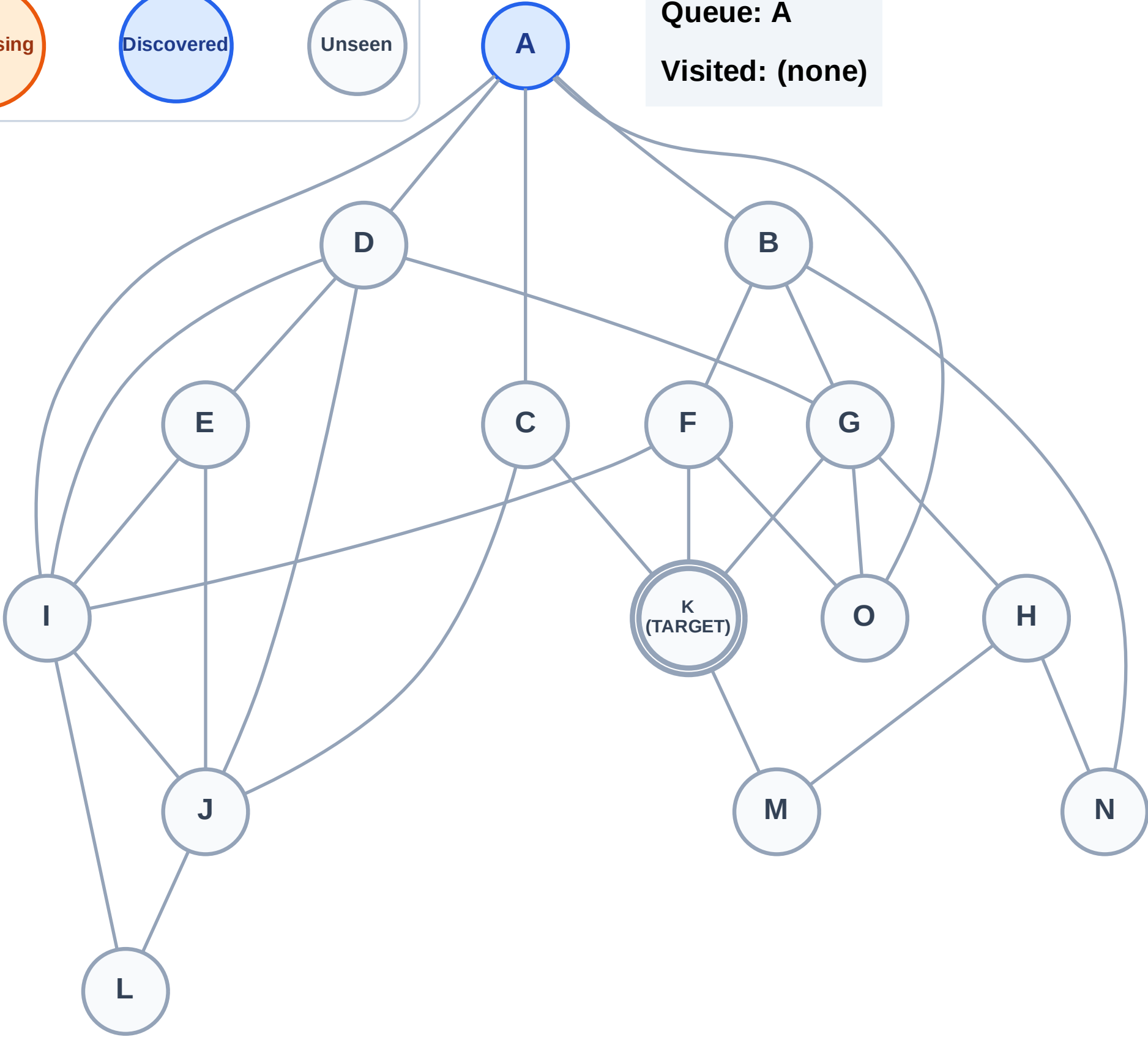
Processing

Discovered

Unseen

Queue: A

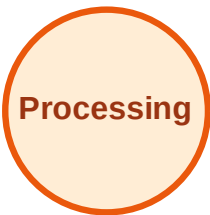
Visited: (none)



BFS Traversal

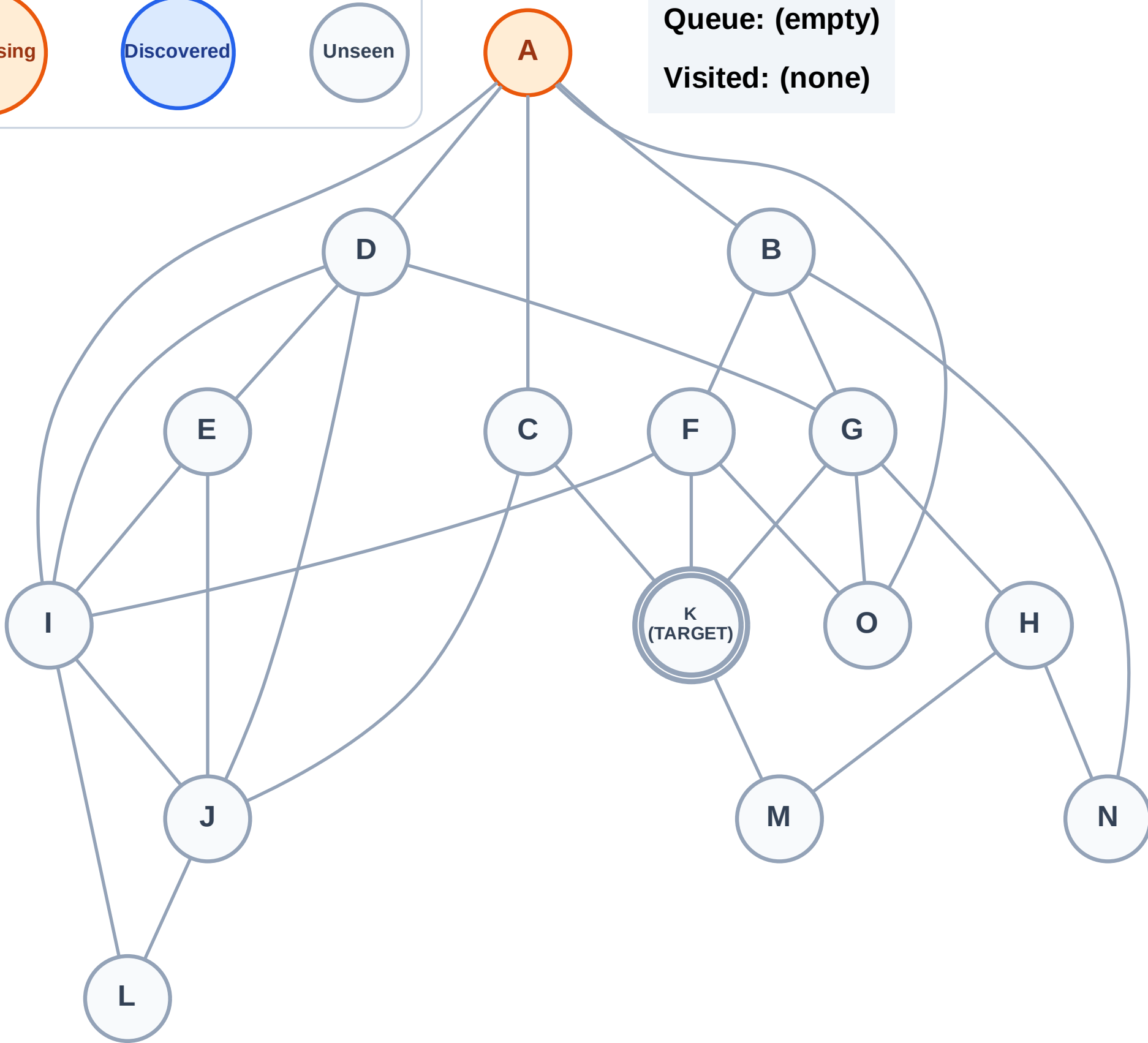
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover B, C, D, I, O from A

Legend

Complete

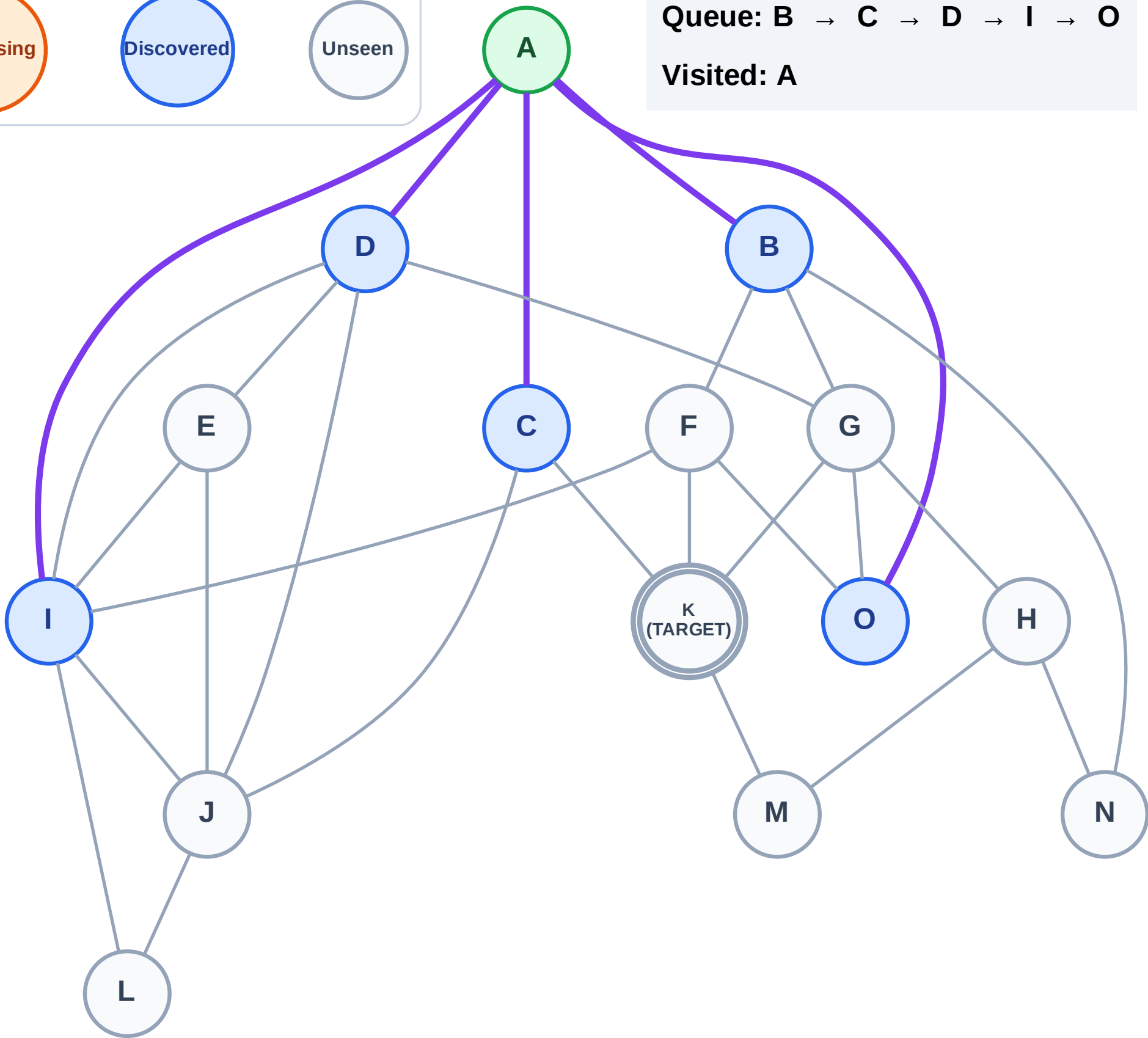
Processing

Discovered

Unseen

Queue: B → C → D → I → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

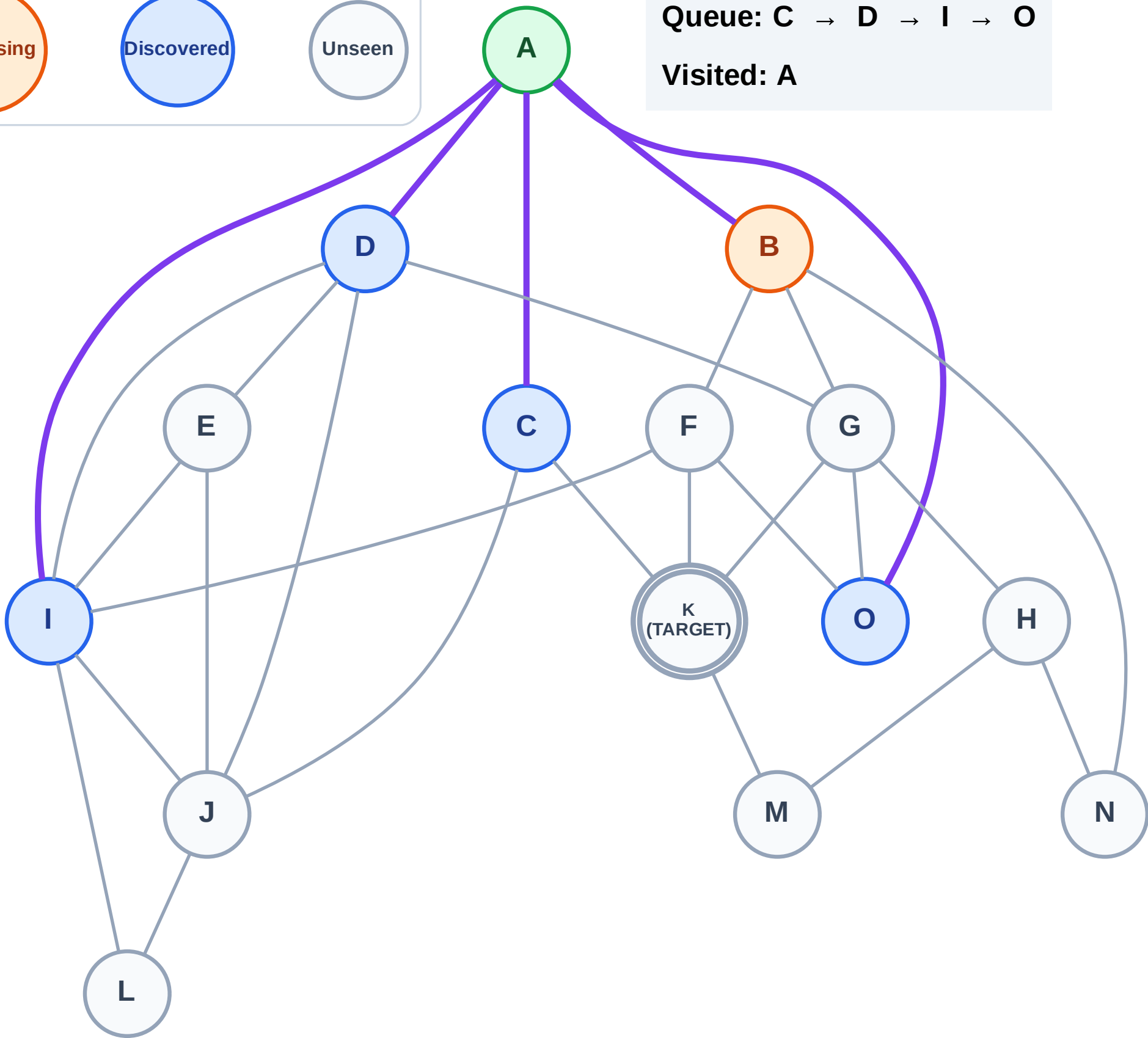
Processing

Discovered

Unseen

Queue: C → D → I → O

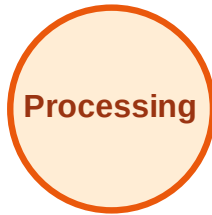
Visited: A



BFS Traversal

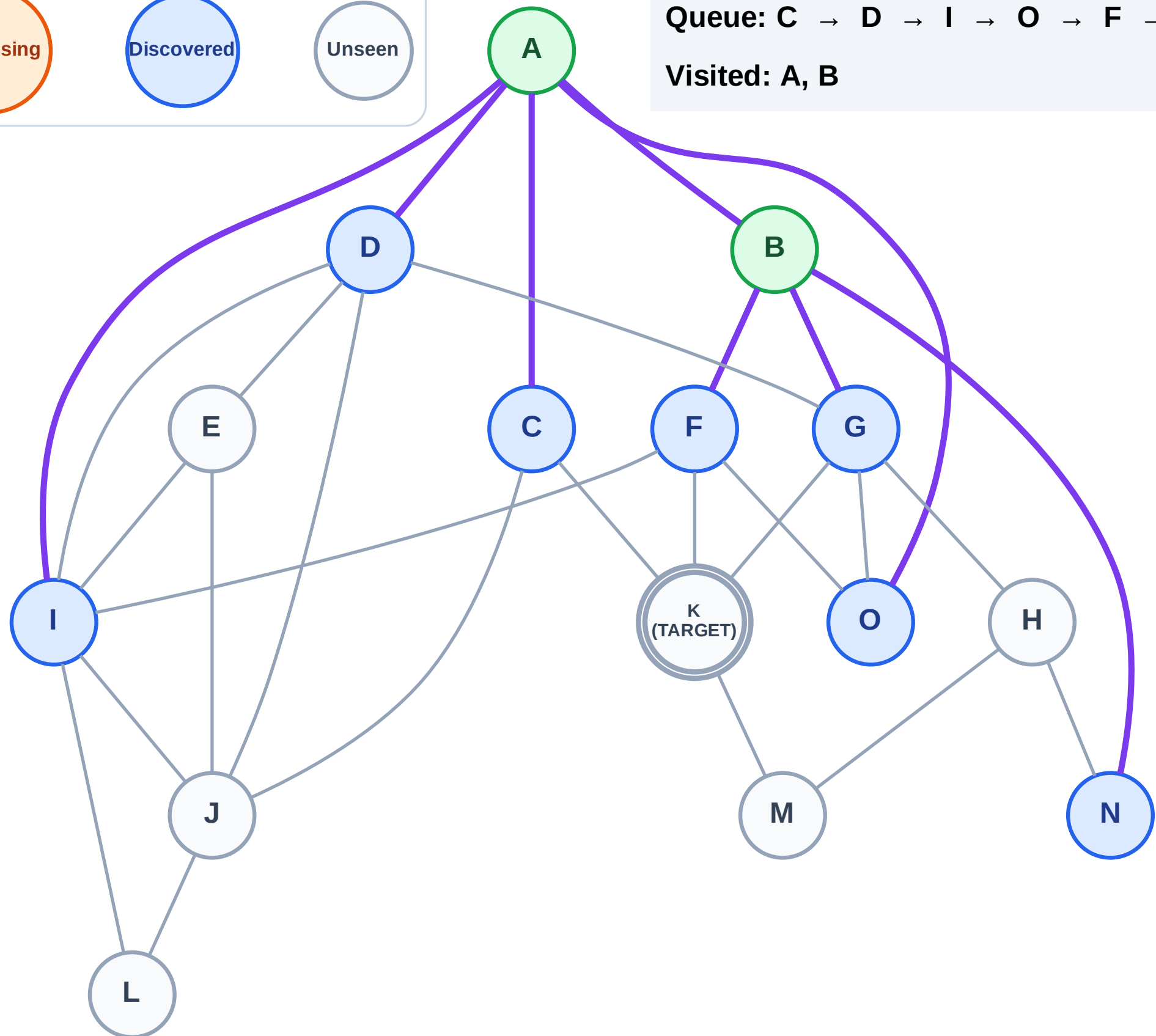
Step 5: Discover F, G, N from B

Legend



Queue: C → D → I → O → F → G → N

Visited: A, B



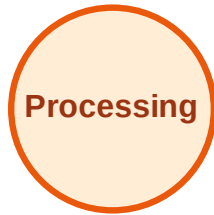
BFS Traversal

Step 6: Process node C

Legend



Complete



Processing

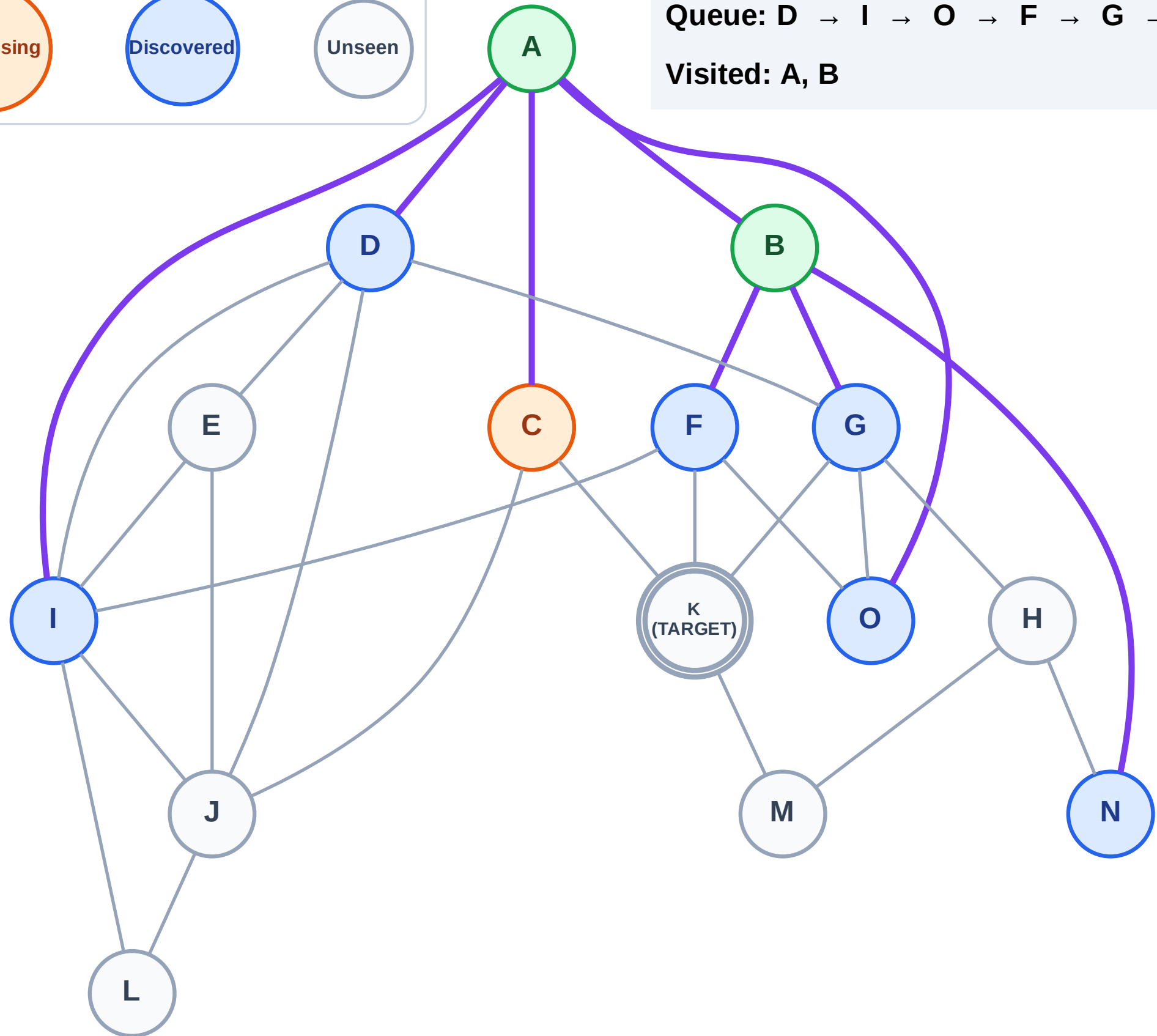


Discovered



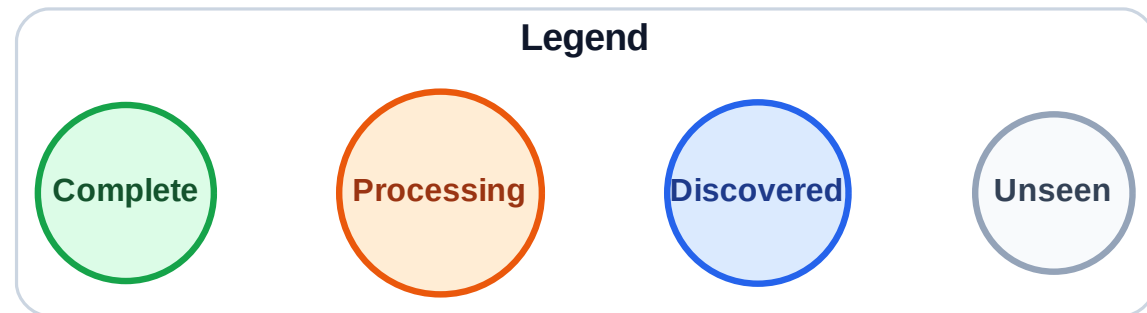
Unseen

Queue: D → I → O → F → G → N
Visited: A, B

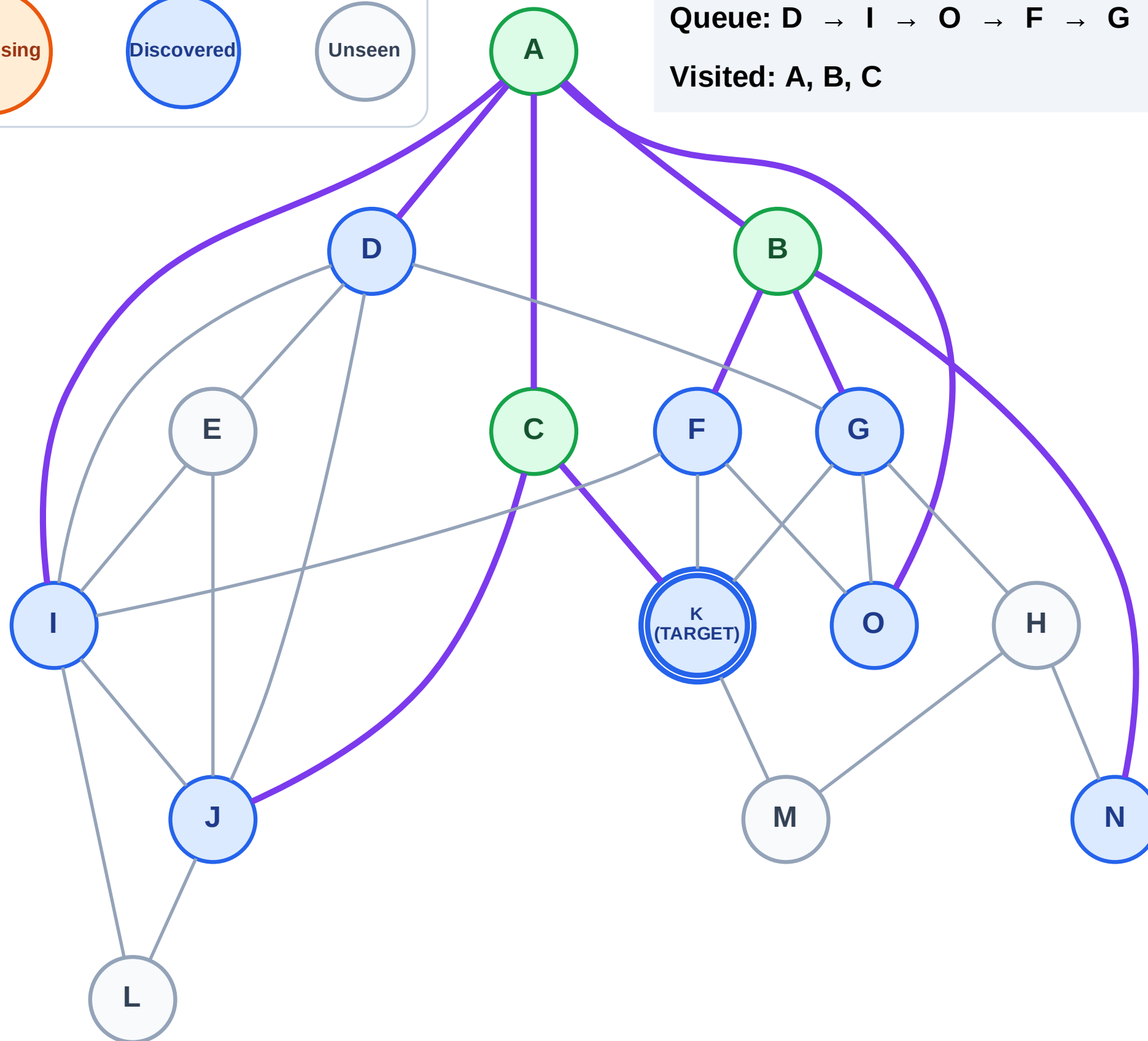


BFS Traversal

Step 7: Discover J, K from C



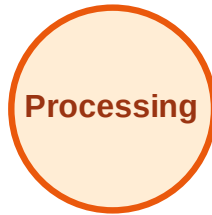
Queue: D → I → O → F → G → N → J → K
Visited: A, B, C



BFS Traversal

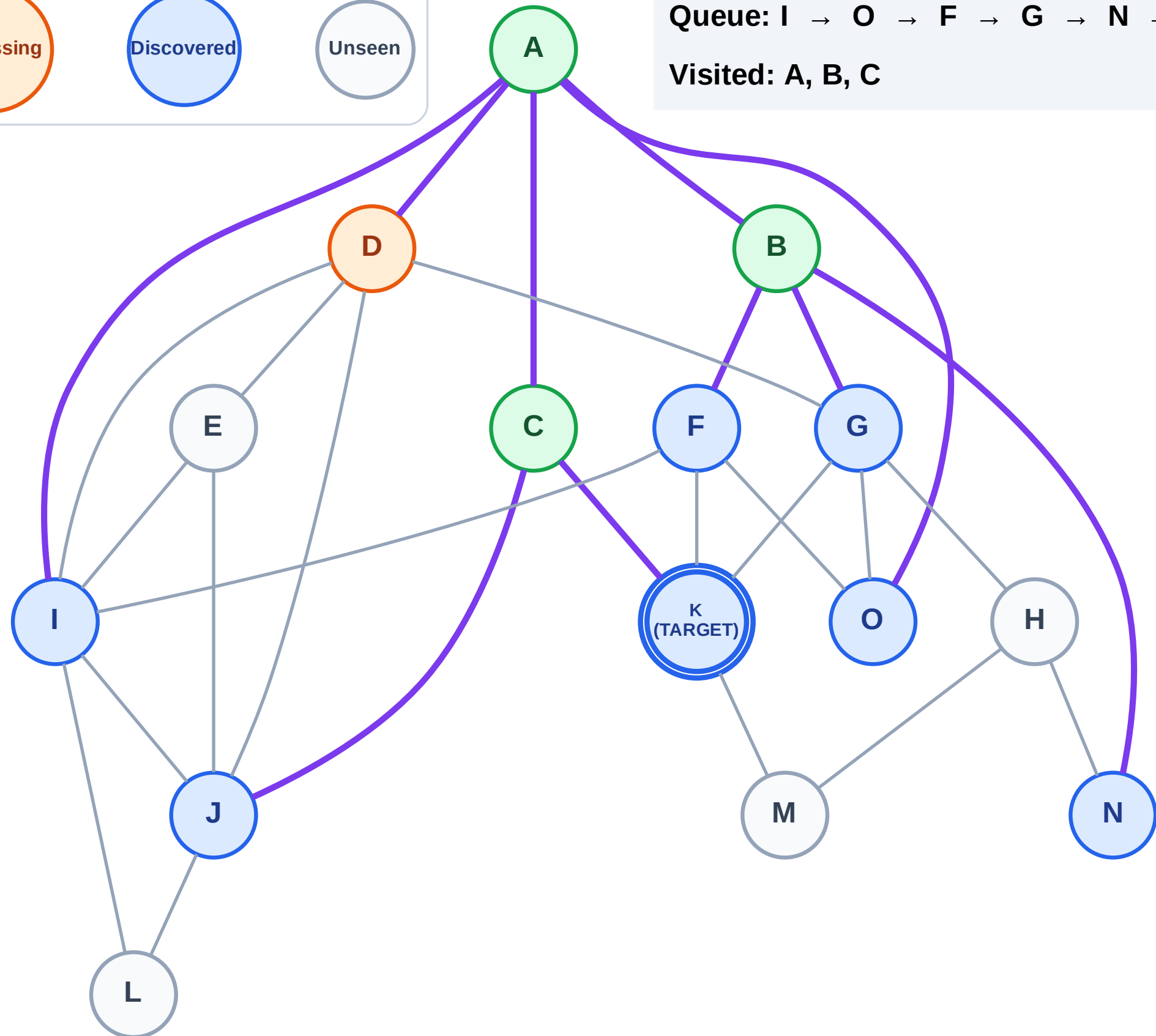
Step 8: Process node D

Legend



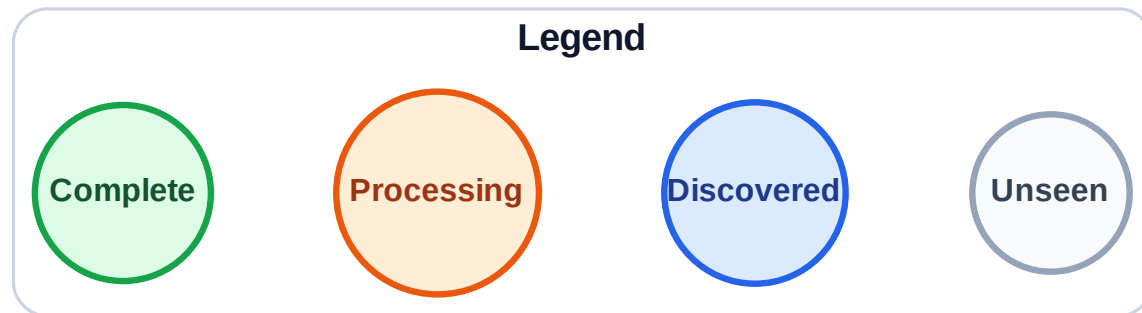
Queue: I → O → F → G → N → J → K

Visited: A, B, C



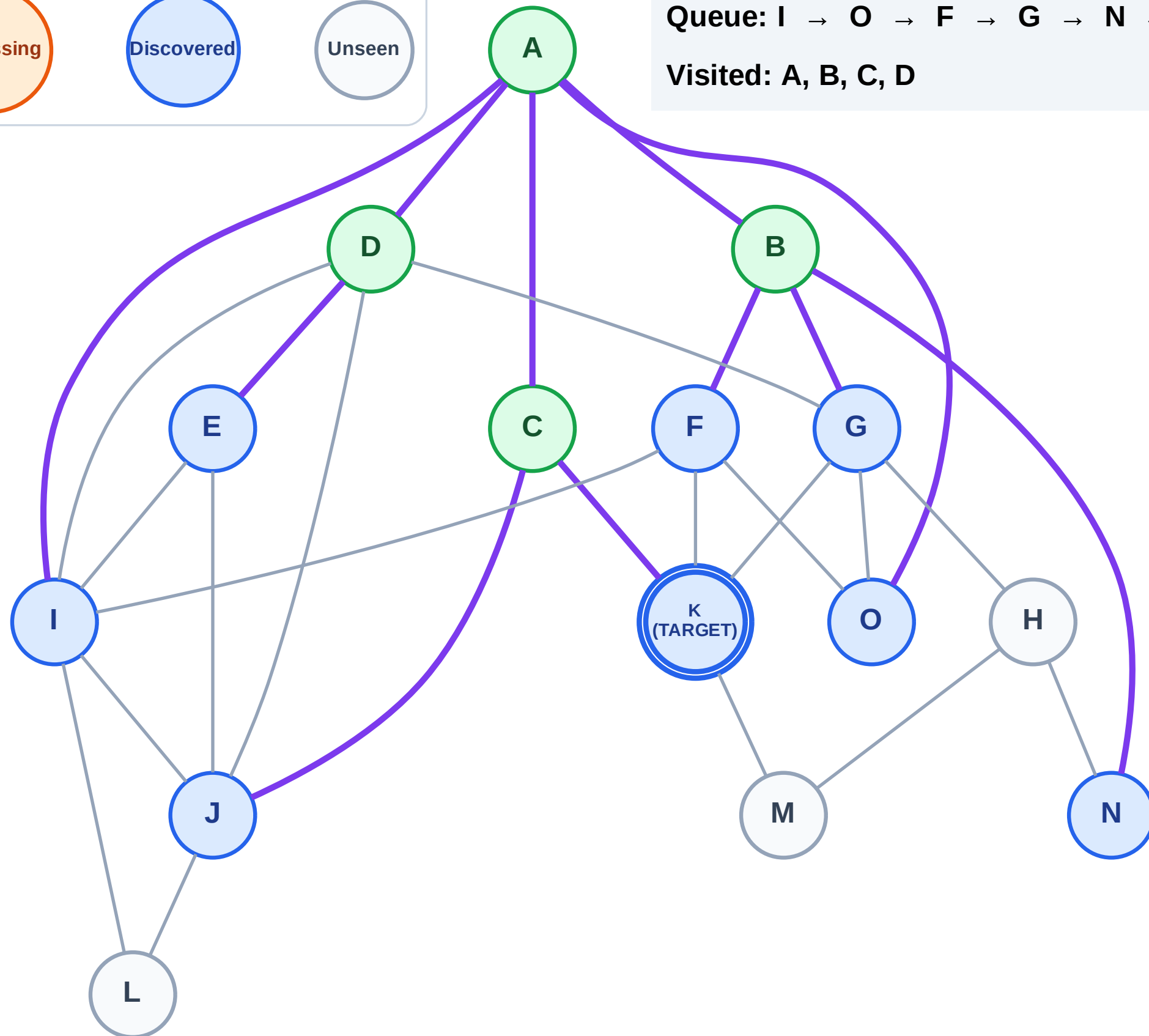
BFS Traversal

Step 9: Discover E from D



Queue: I → O → F → G → N → J → K → E

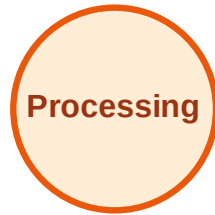
Visited: A, B, C, D



BFS Traversal

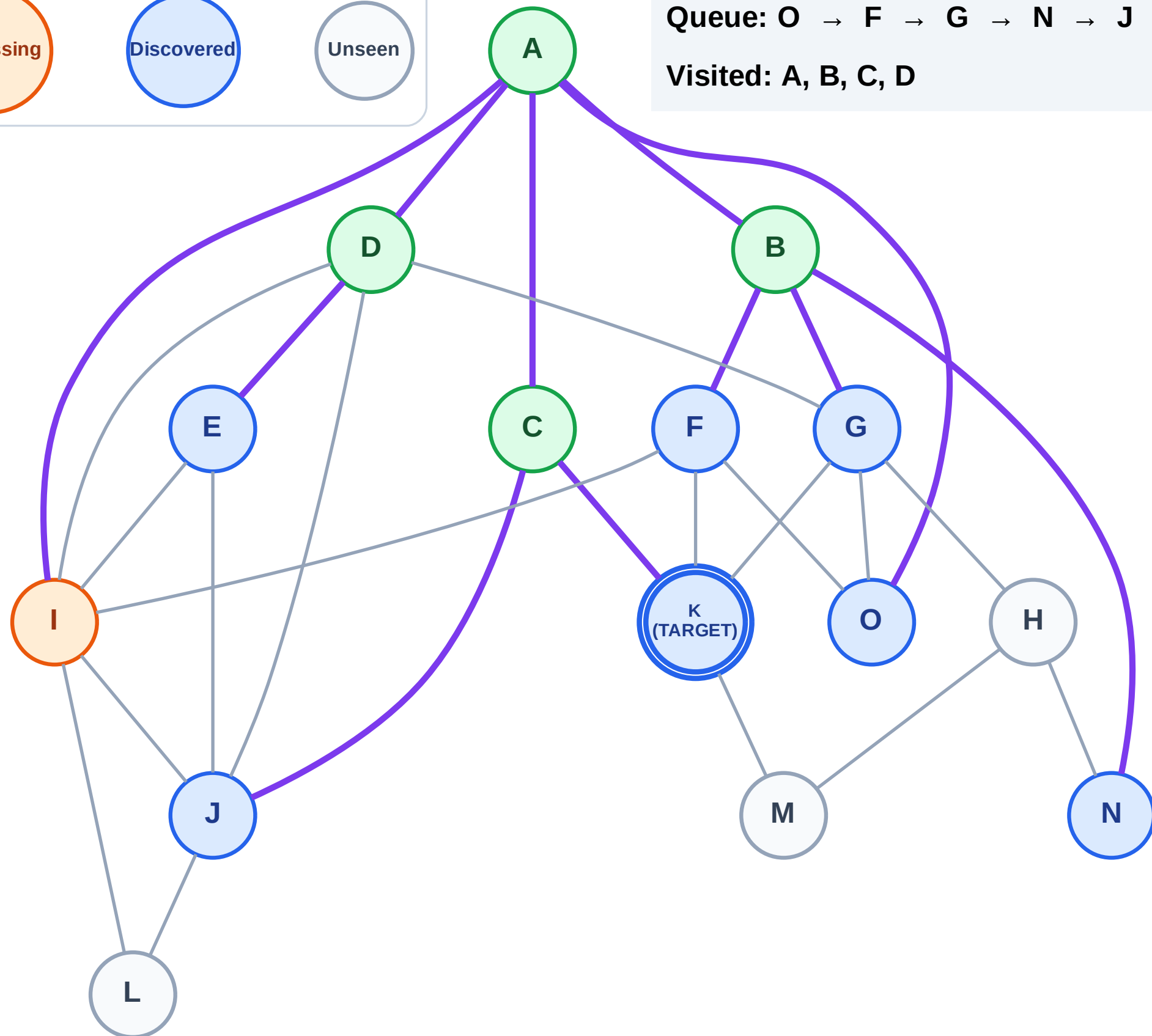
Step 10: Process node I

Legend



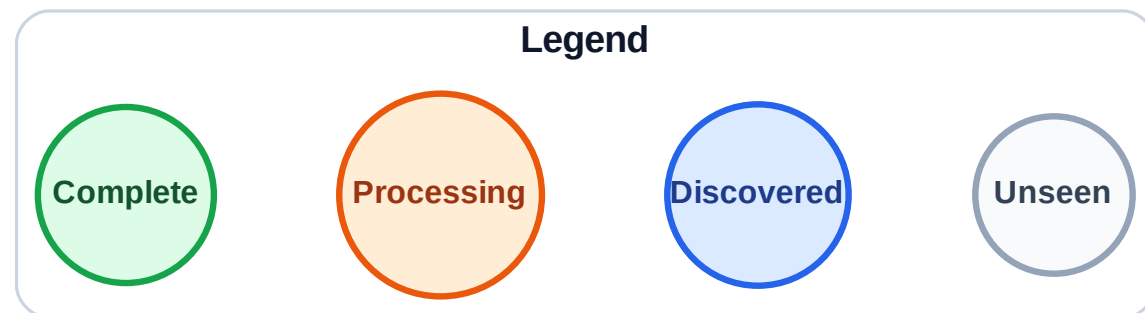
Queue: O → F → G → N → J → K → E

Visited: A, B, C, D



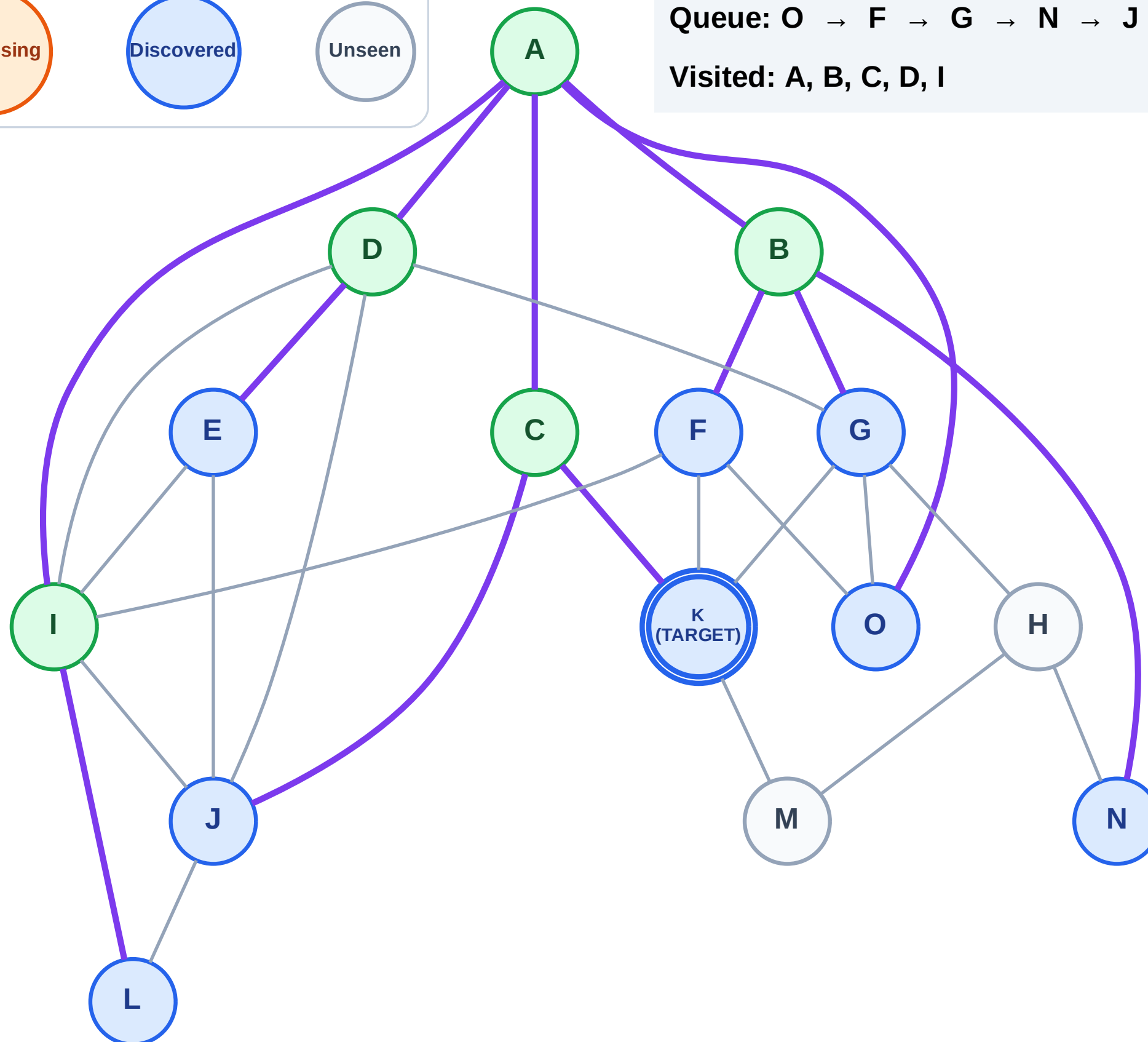
BFS Traversal

Step 11: Discover L from I



Queue: O → F → G → N → J → K → E → L

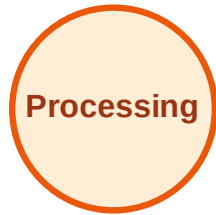
Visited: A, B, C, D, I



BFS Traversal

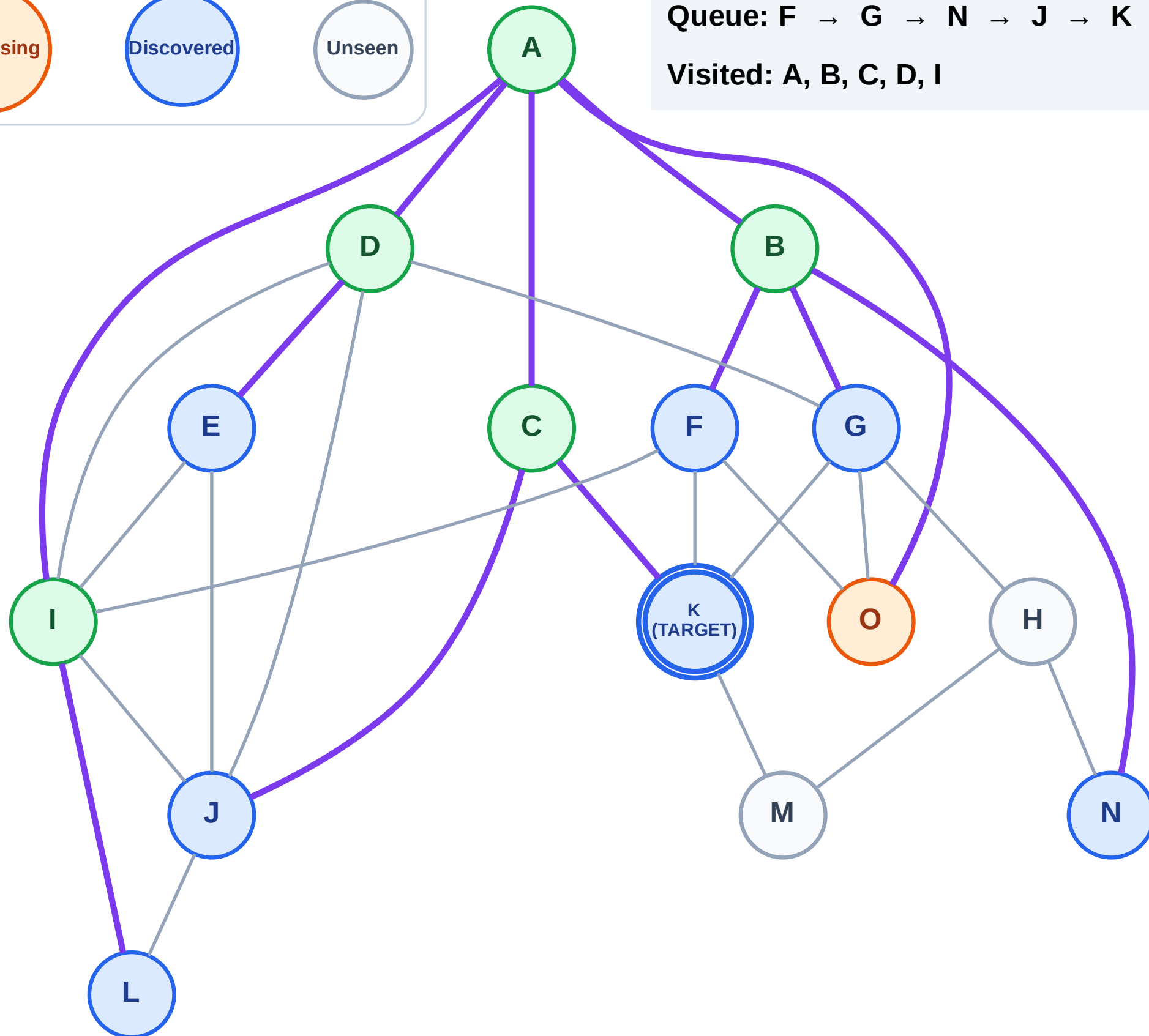
Step 12: Process node O

Legend



Queue: F → G → N → J → K → E → L

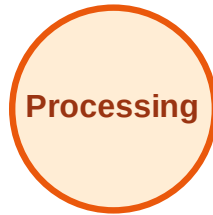
Visited: A, B, C, D, I



BFS Traversal

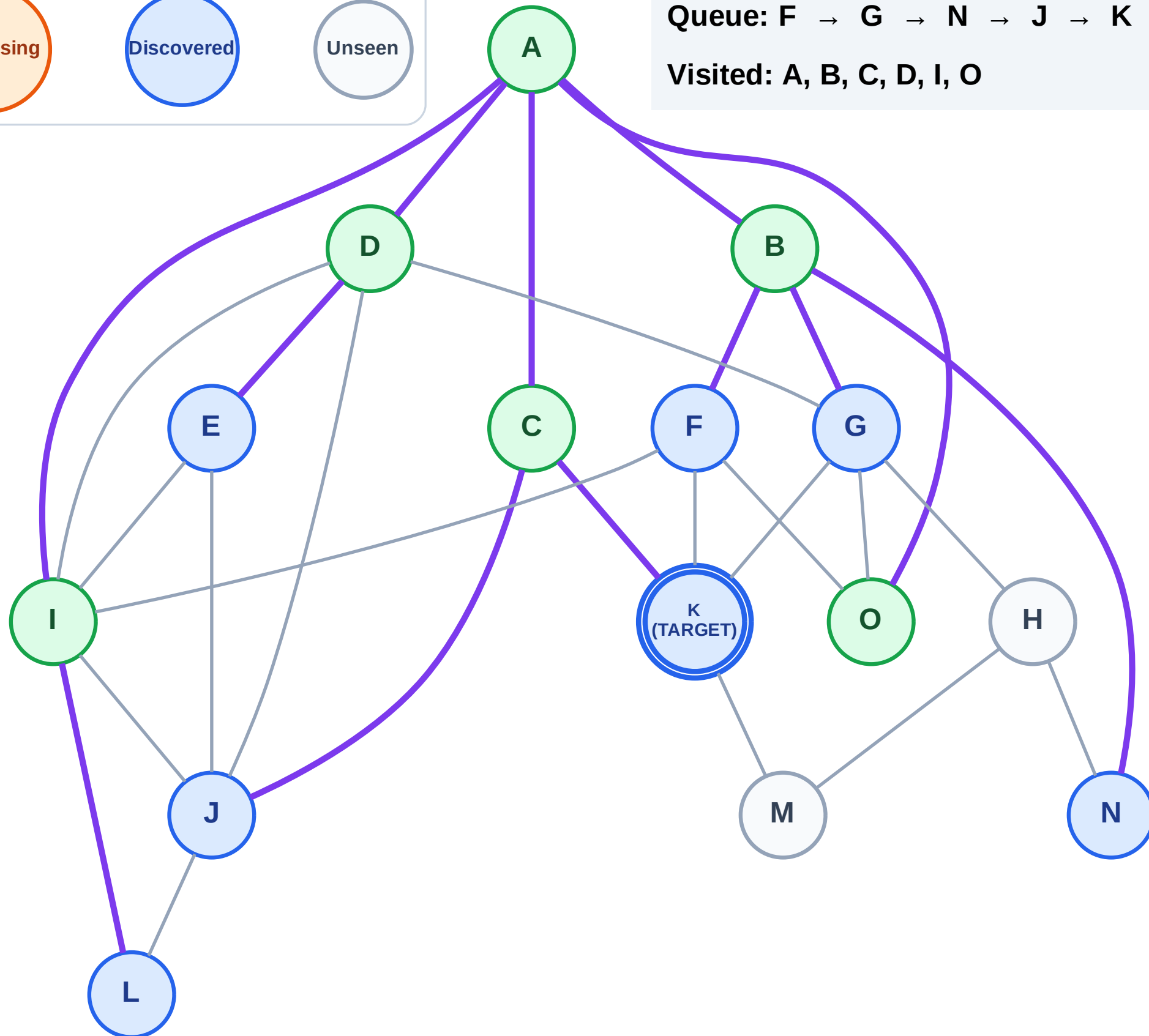
Step 13: No new neighbors from O

Legend



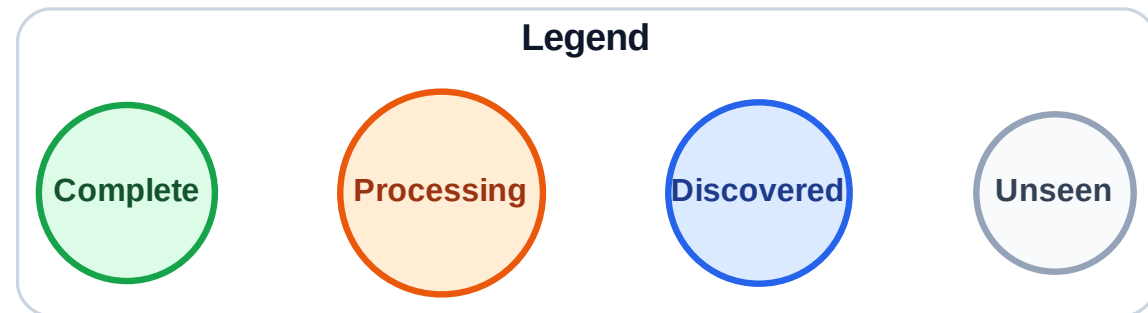
Queue: F → G → N → J → K → E → L

Visited: A, B, C, D, I, O



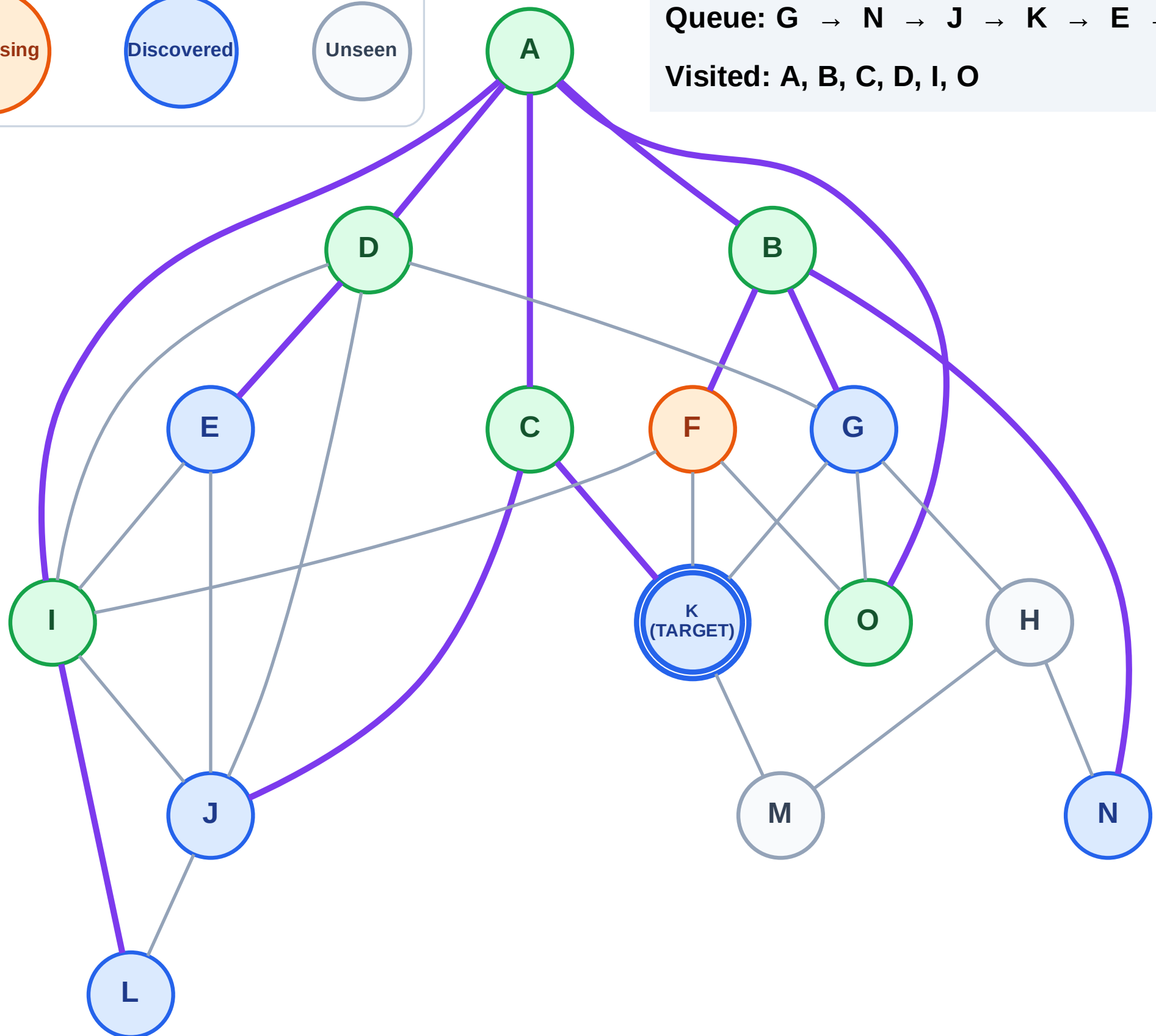
BFS Traversal

Step 14: Process node F



Queue: G → N → J → K → E → L

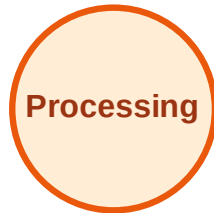
Visited: A, B, C, D, I, O



BFS Traversal

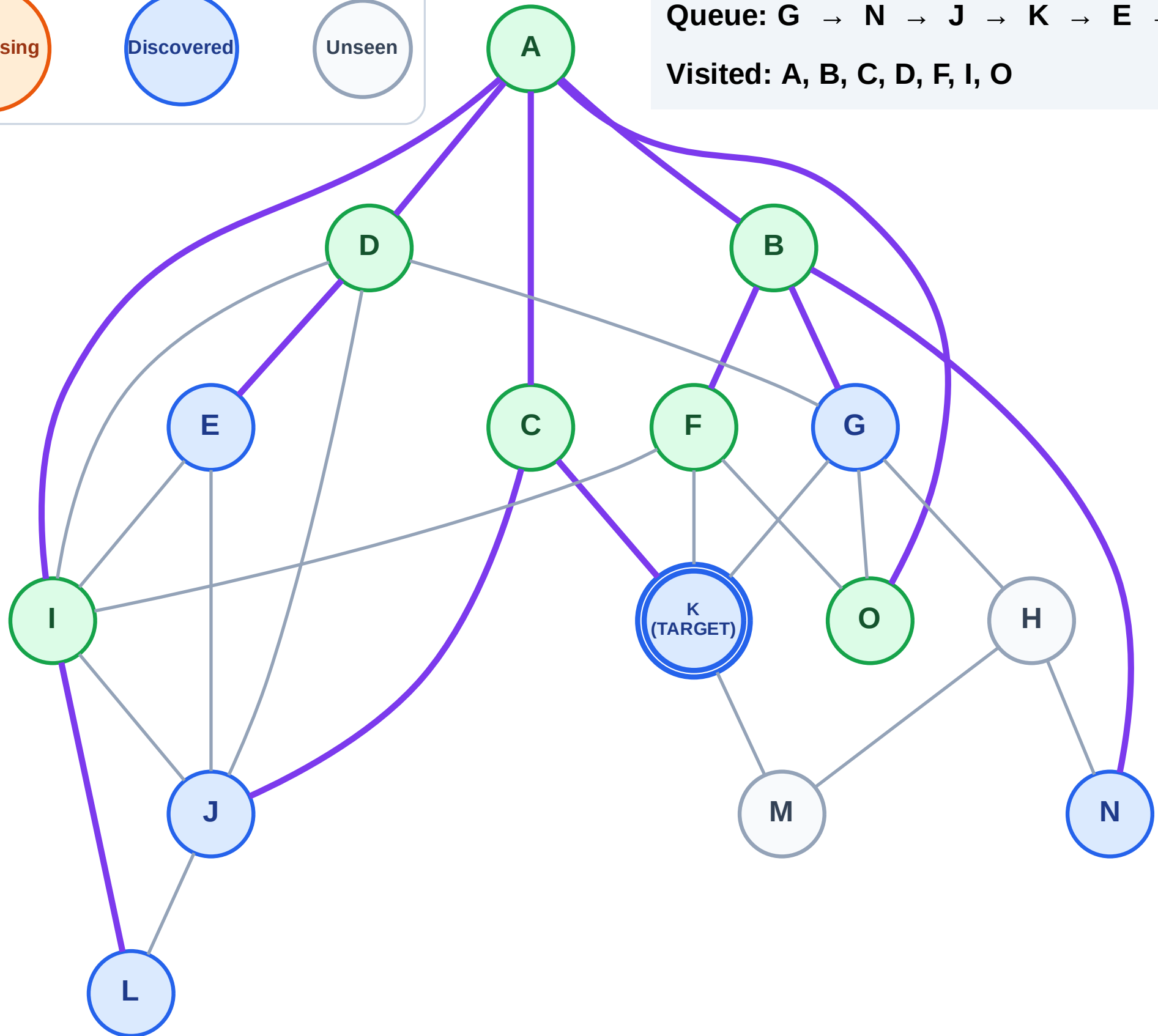
Step 15: No new neighbors from F

Legend



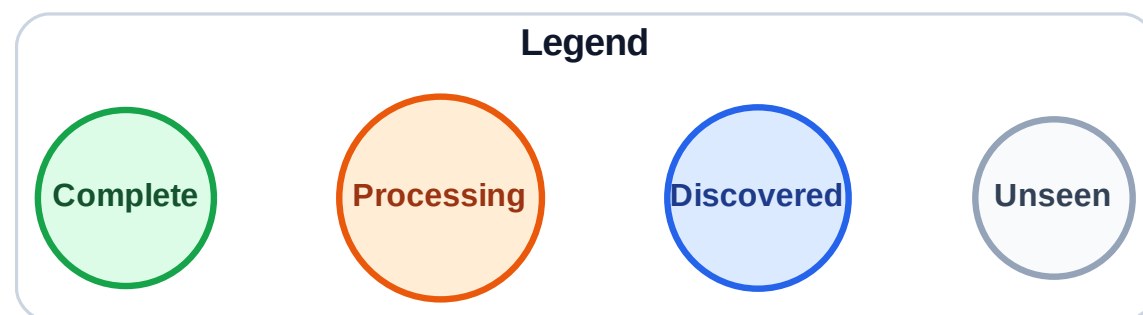
Queue: G → N → J → K → E → L

Visited: A, B, C, D, F, I, O

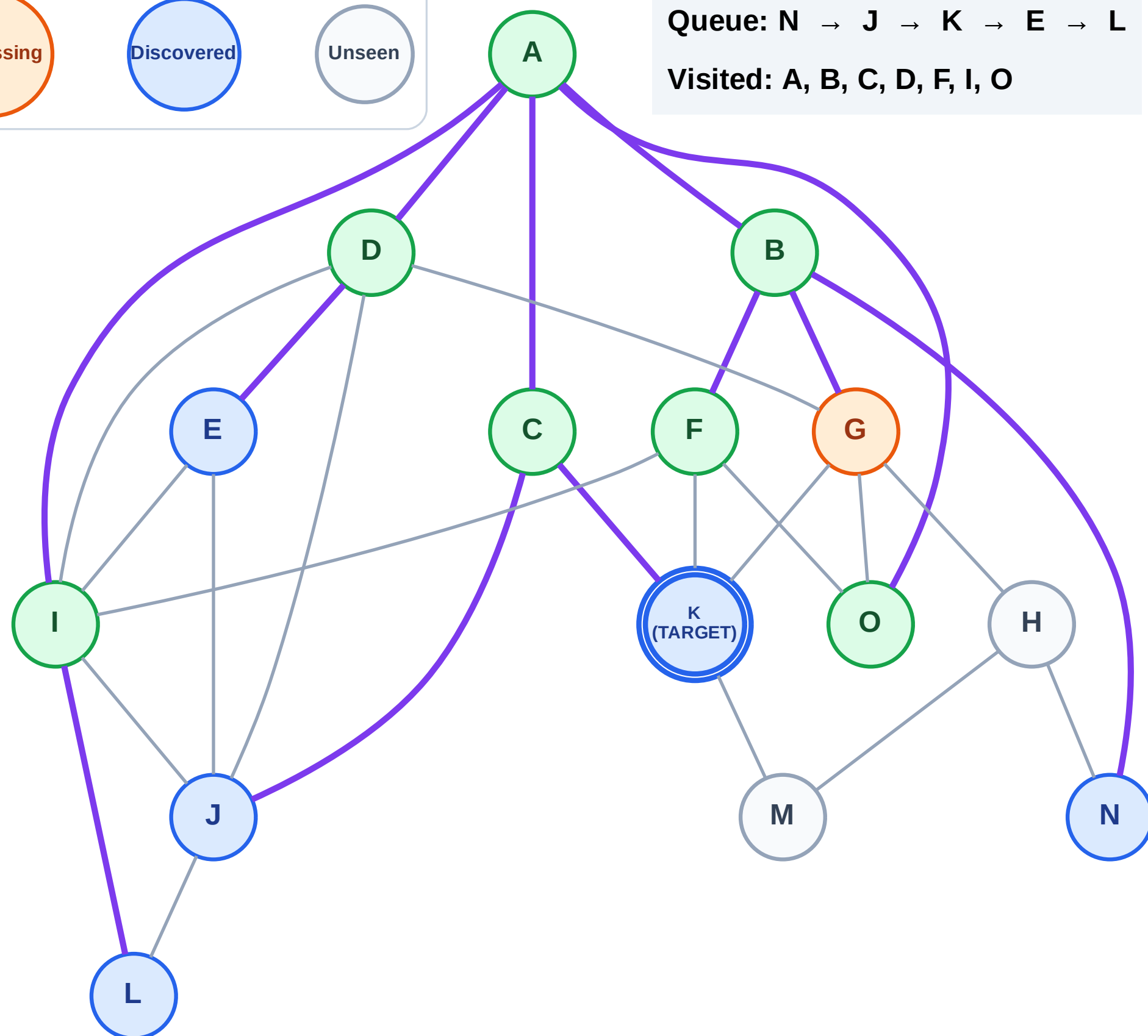


BFS Traversal

Step 16: Process node G

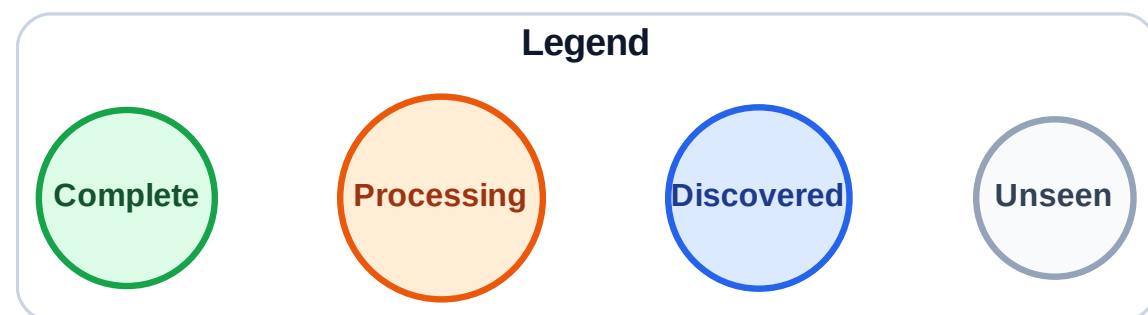


Queue: N → J → K → E → L
Visited: A, B, C, D, F, I, O



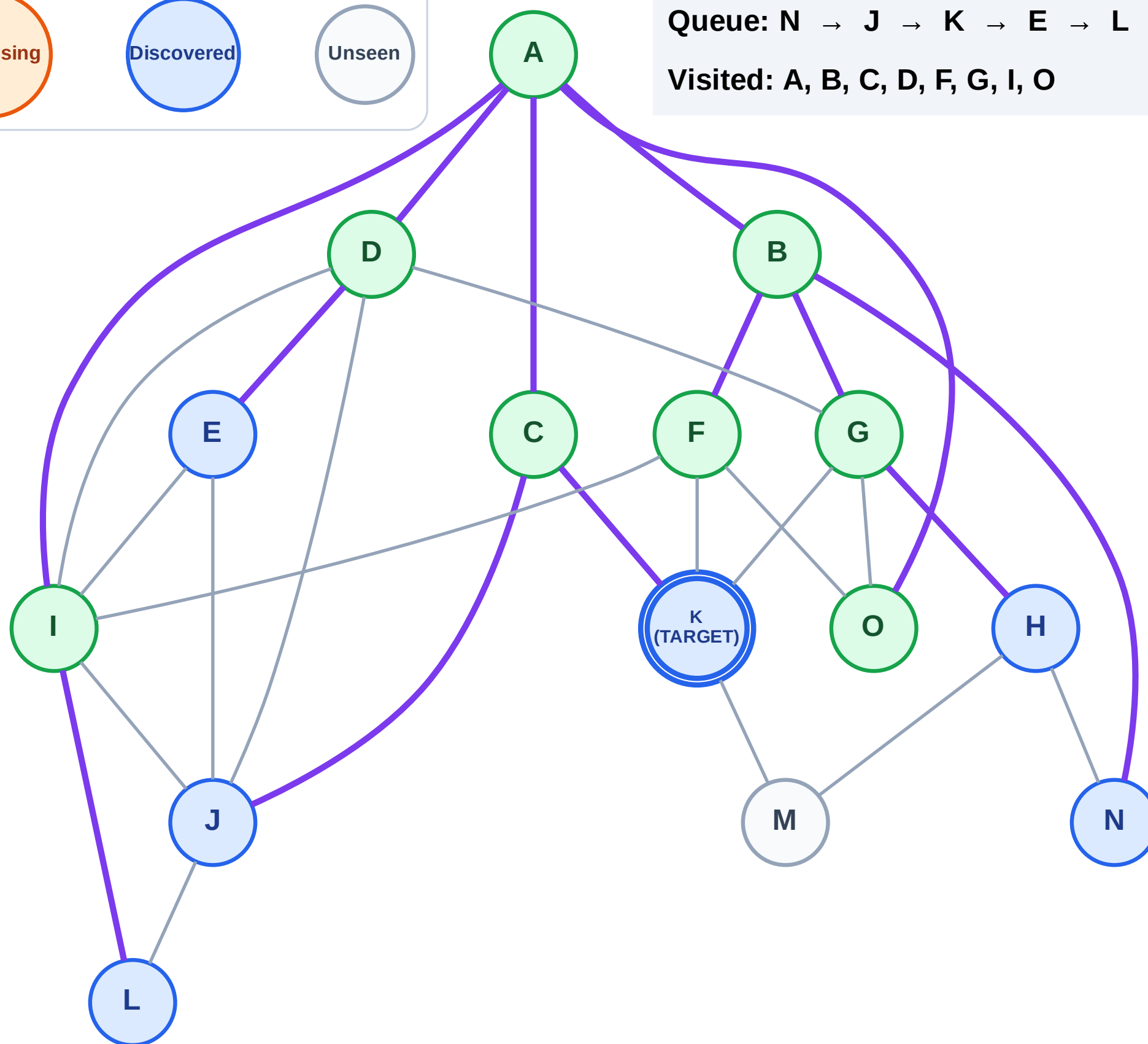
BFS Traversal

Step 17: Discover H from G



Queue: N → J → K → E → L → H

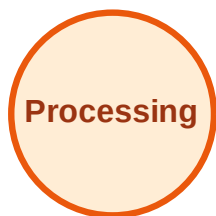
Visited: A, B, C, D, F, G, I, O



BFS Traversal

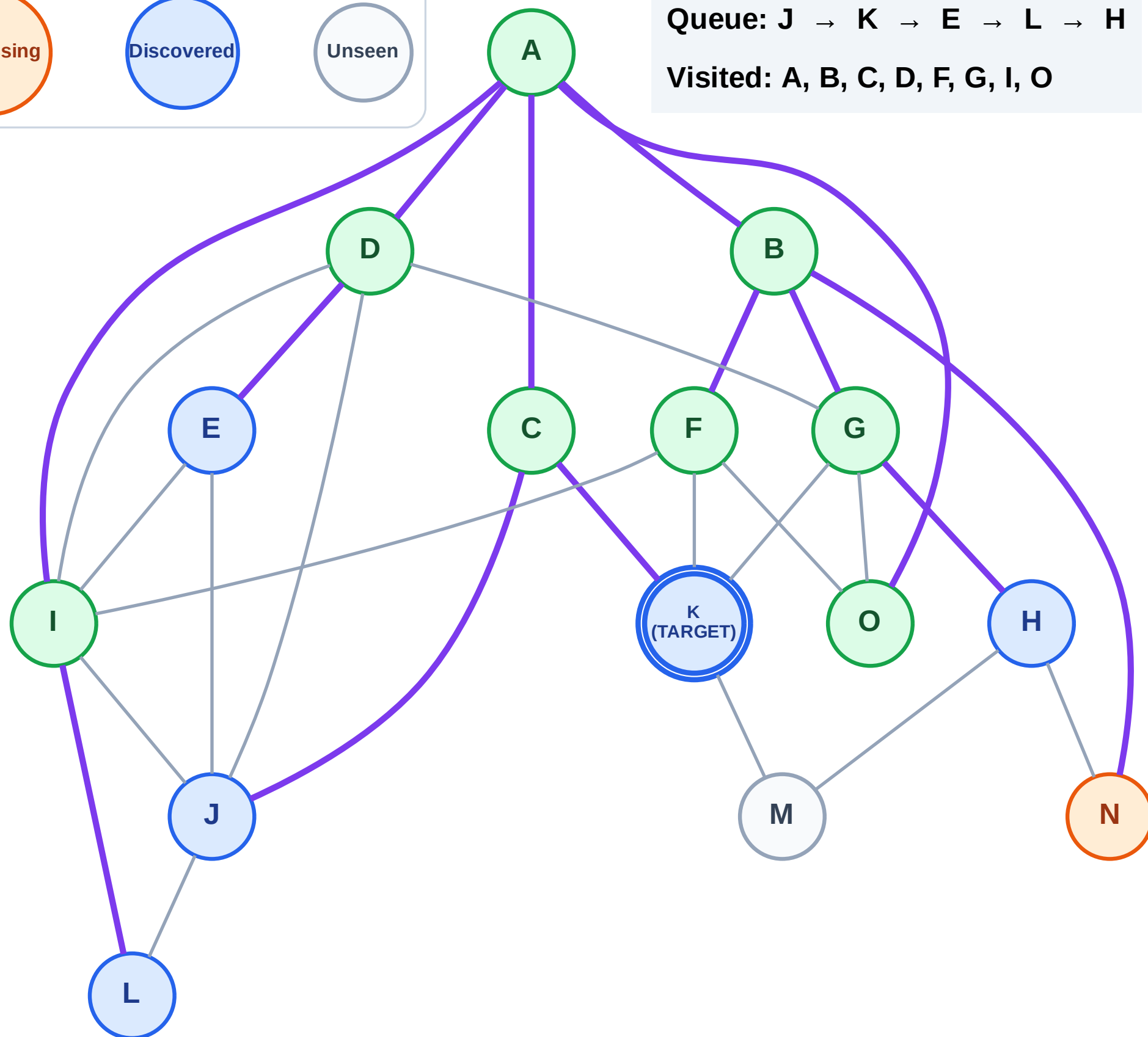
Step 18: Process node N

Legend



Queue: J → K → E → L → H

Visited: A, B, C, D, F, G, I, O



BFS Traversal

Step 19: No new neighbors from N

Legend

Complete

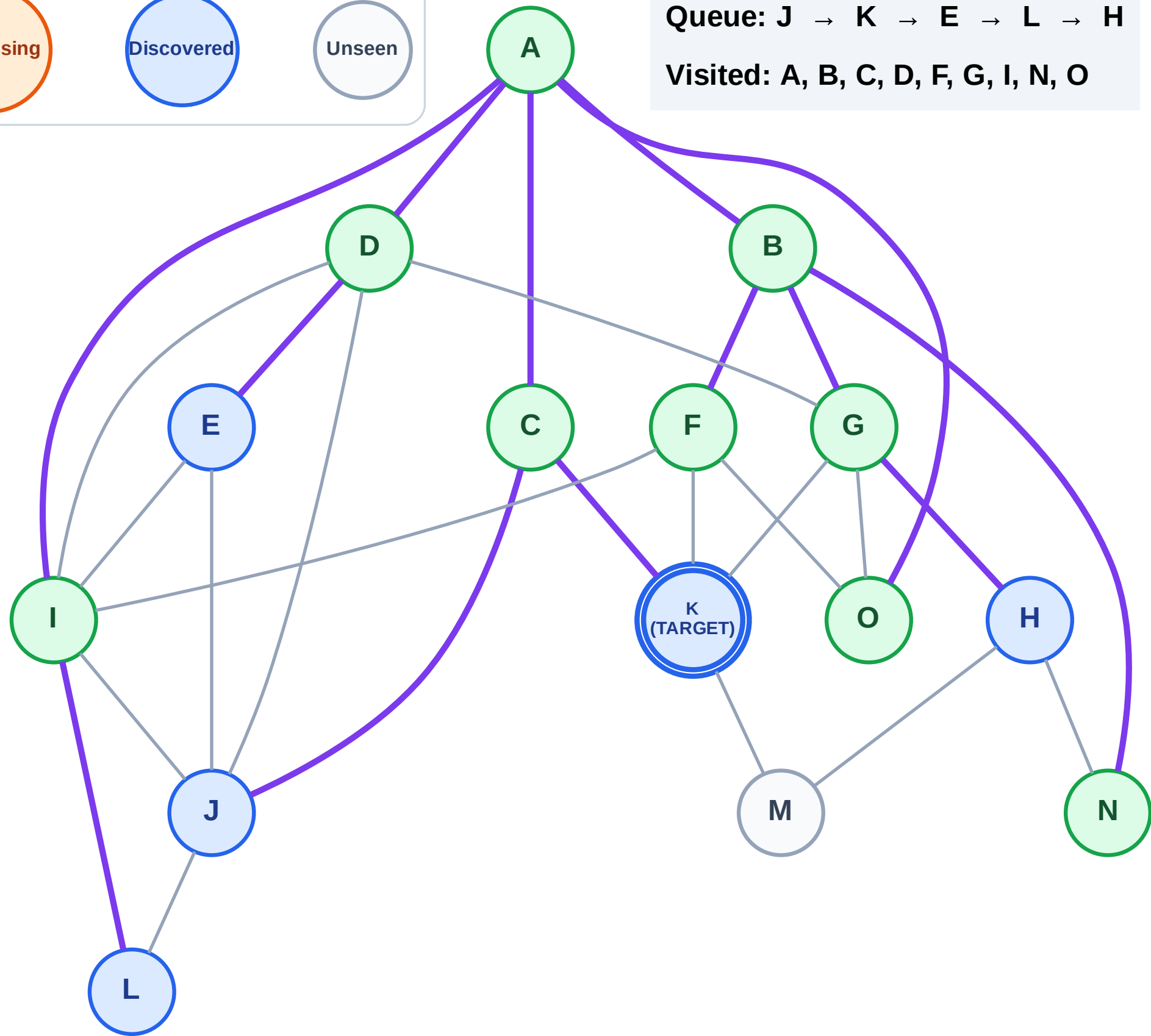
Processing

Discovered

Unseen

Queue: J → K → E → L → H

Visited: A, B, C, D, F, G, I, N, O



BFS Traversal

Step 20: Process node J

Legend

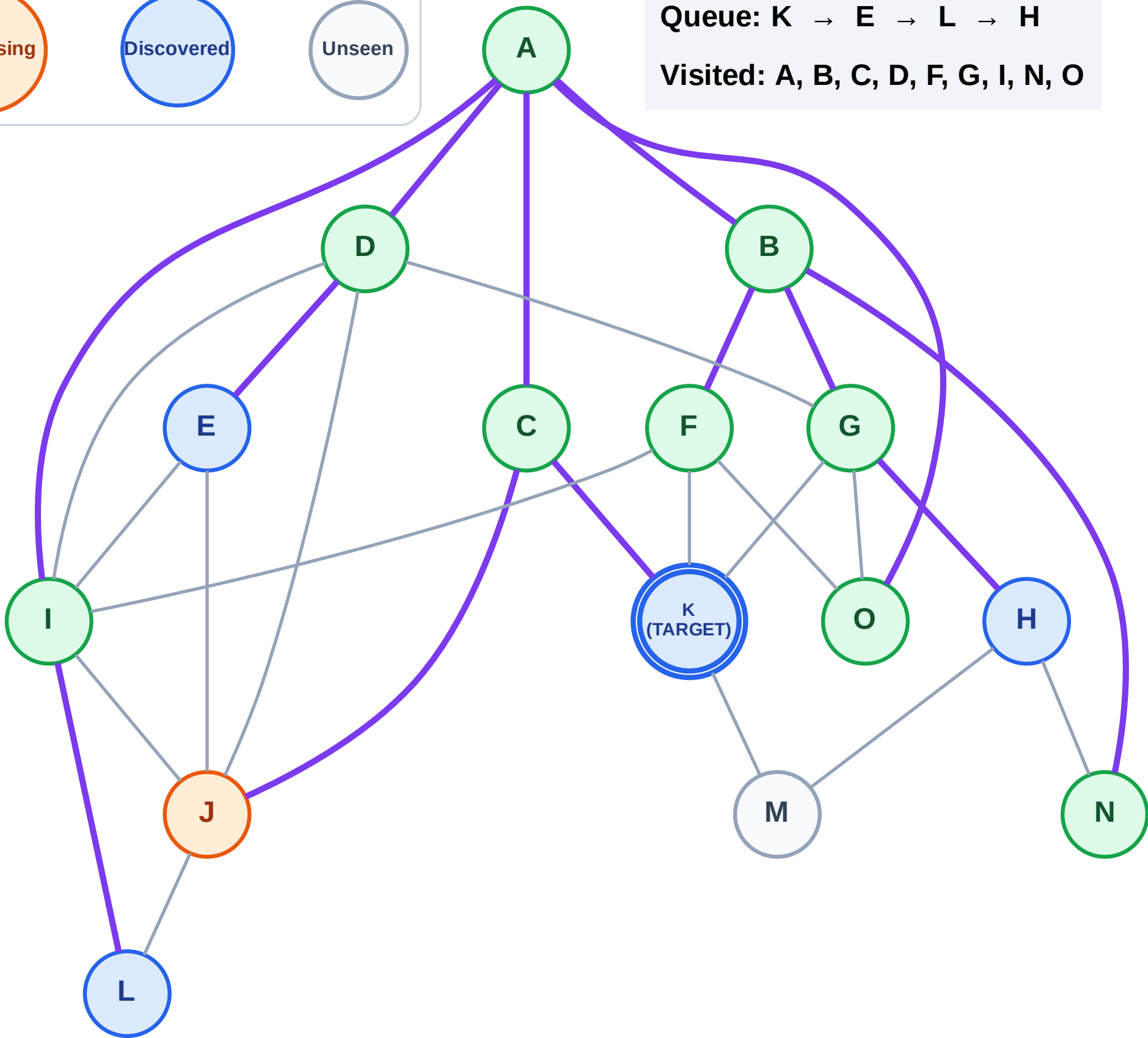
Complete

Processing

Discovered

Unseen

Queue: K → E → L → H
Visited: A, B, C, D, F, G, I, N, O



BFS Traversal

Step 21: No new neighbors from J

Legend

Complete

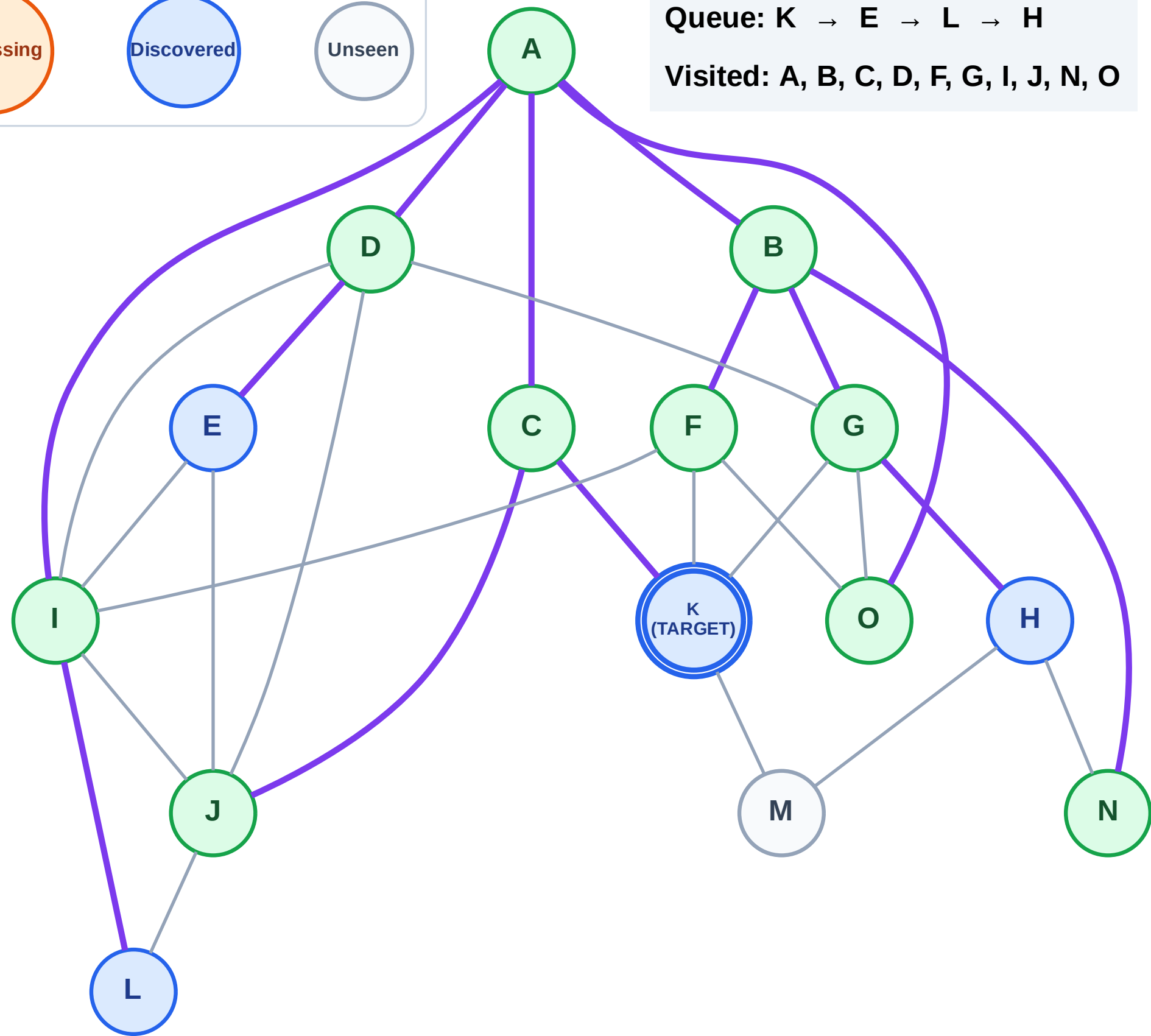
Processing

Discovered

Unseen

Queue: K → E → L → H

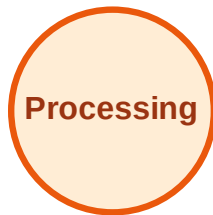
Visited: A, B, C, D, F, G, I, J, N, O



BFS Traversal

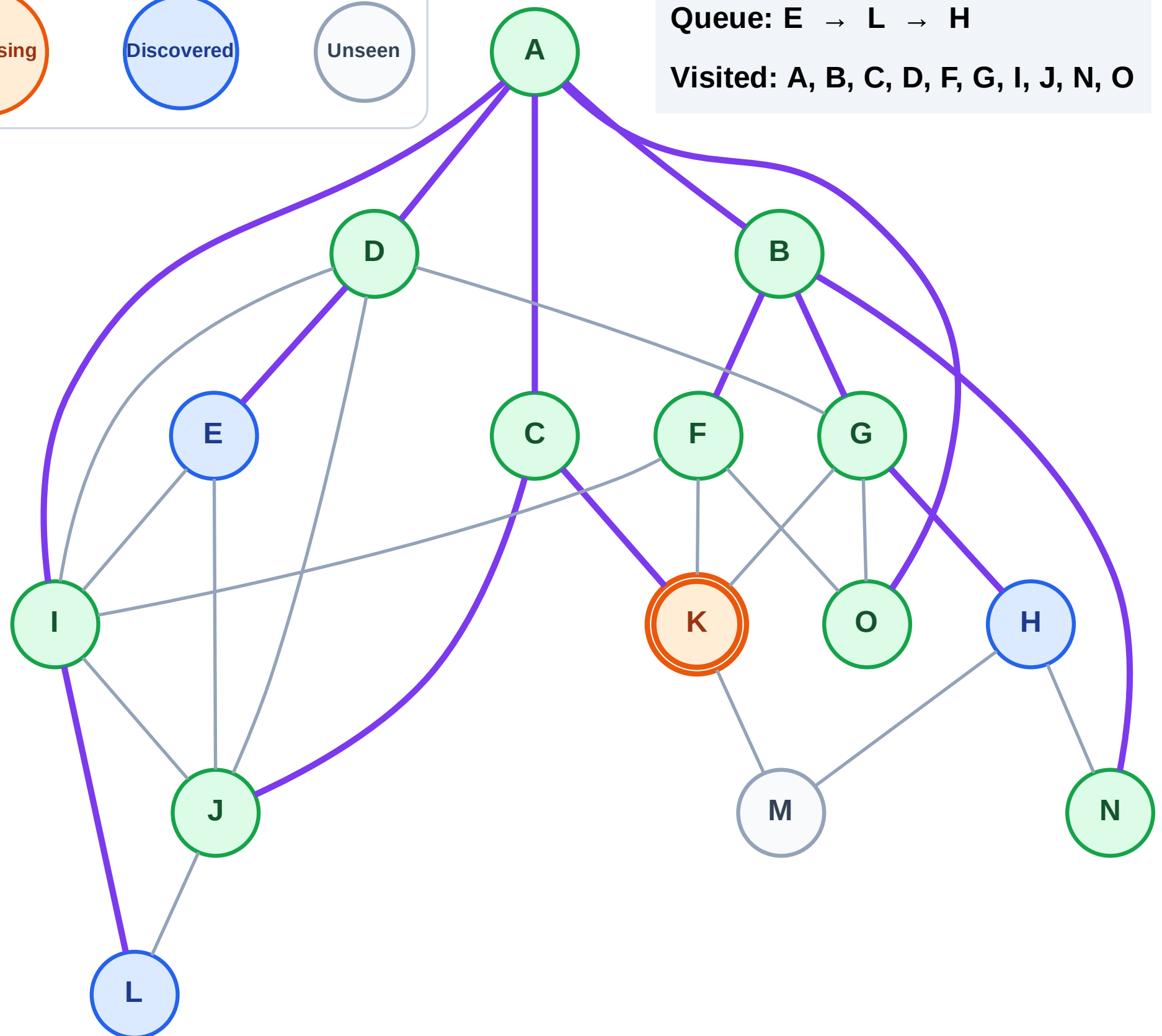
Step 22: Process node K

Legend



Queue: E → L → H

Visited: A, B, C, D, F, G, I, J, N, O



BFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → C → K

Legend

Complete

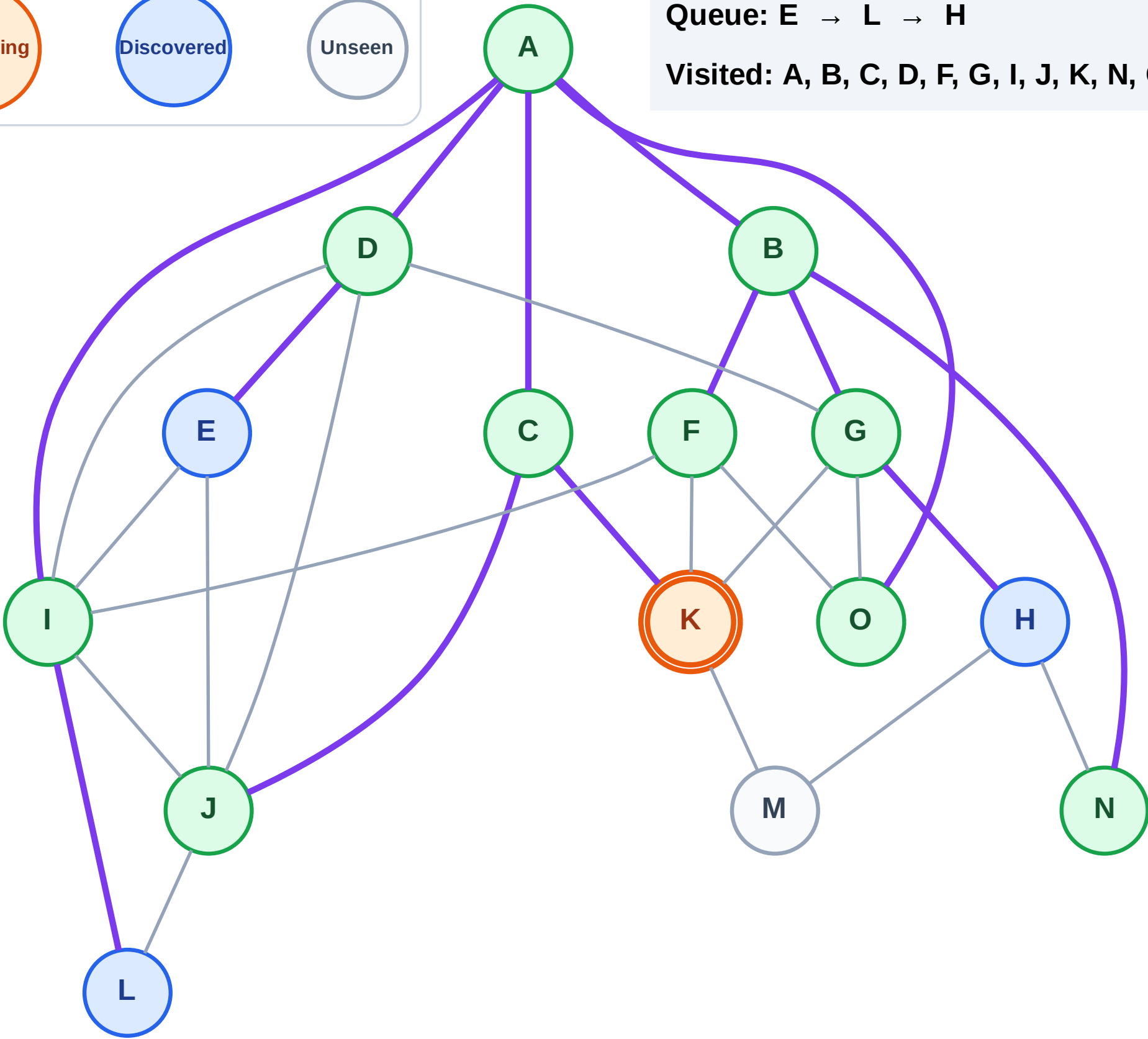
Processing

Discovered

Unseen

Queue: E → L → H

Visited: A, B, C, D, F, G, I, J, K, N, O



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

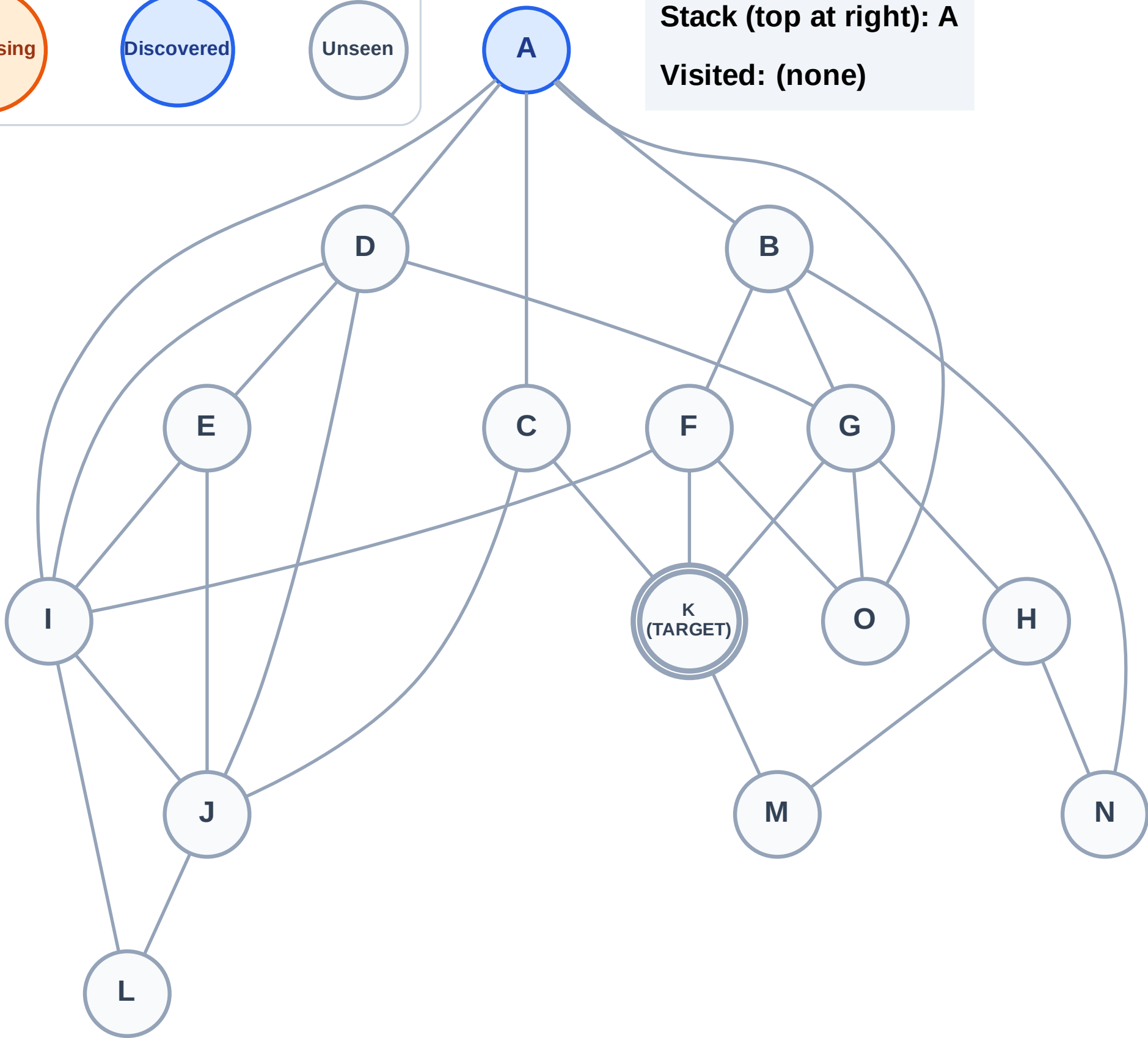
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

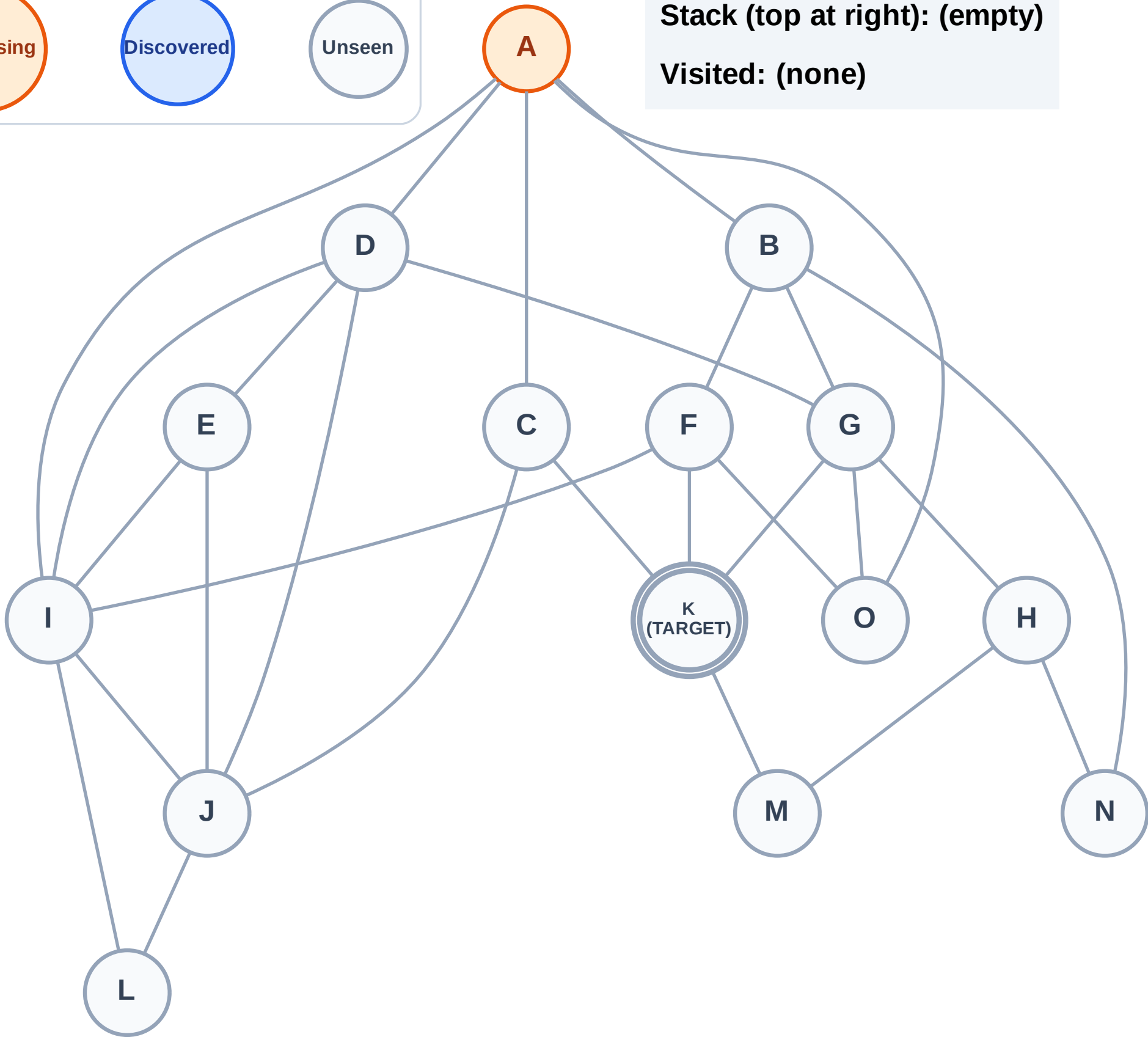
Complete

Processing

Discovered

Unseen

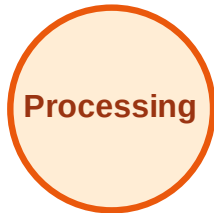
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

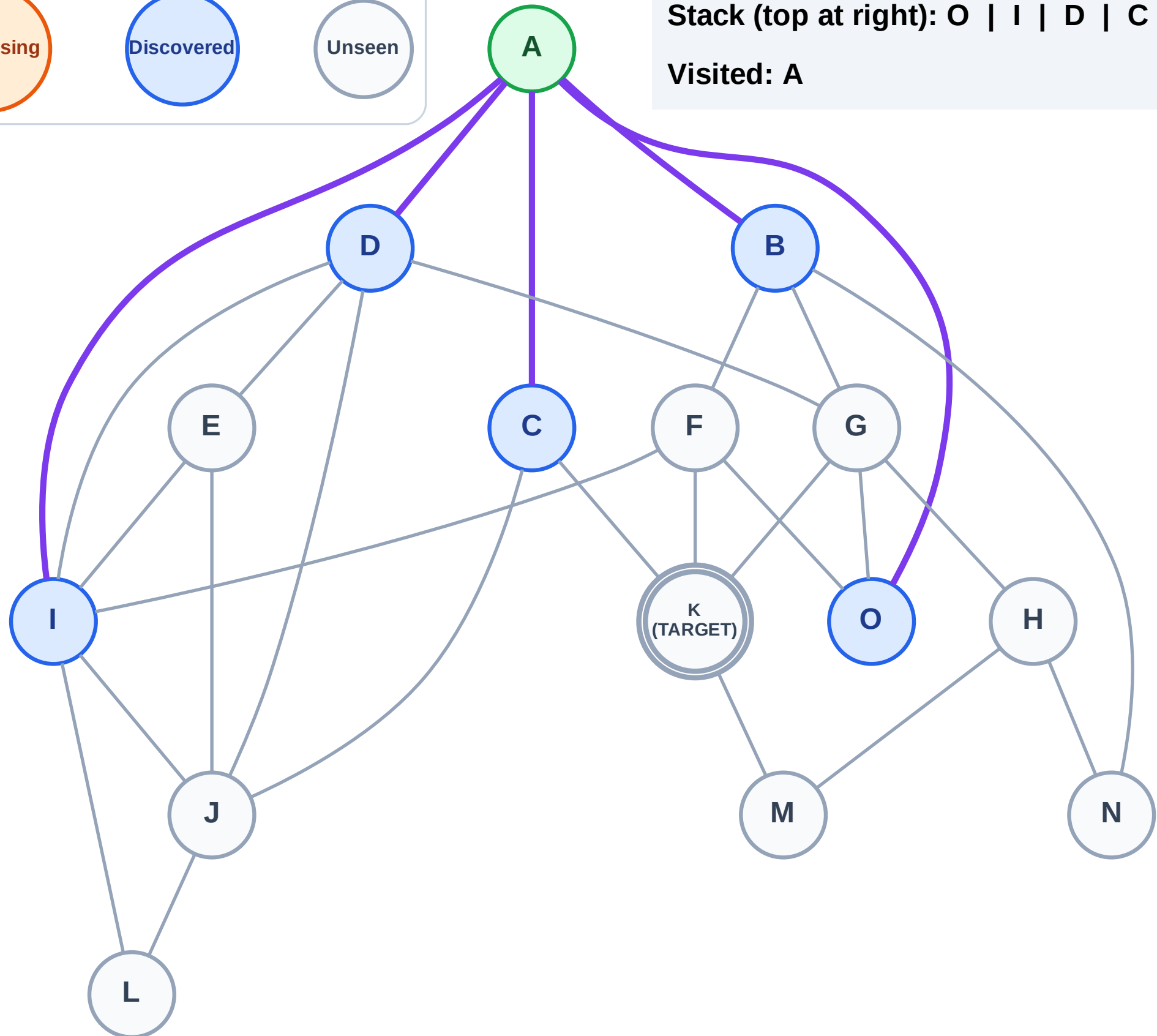
Step 3: Discover O, I, D, C, B from A

Legend



Stack (top at right): O | I | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

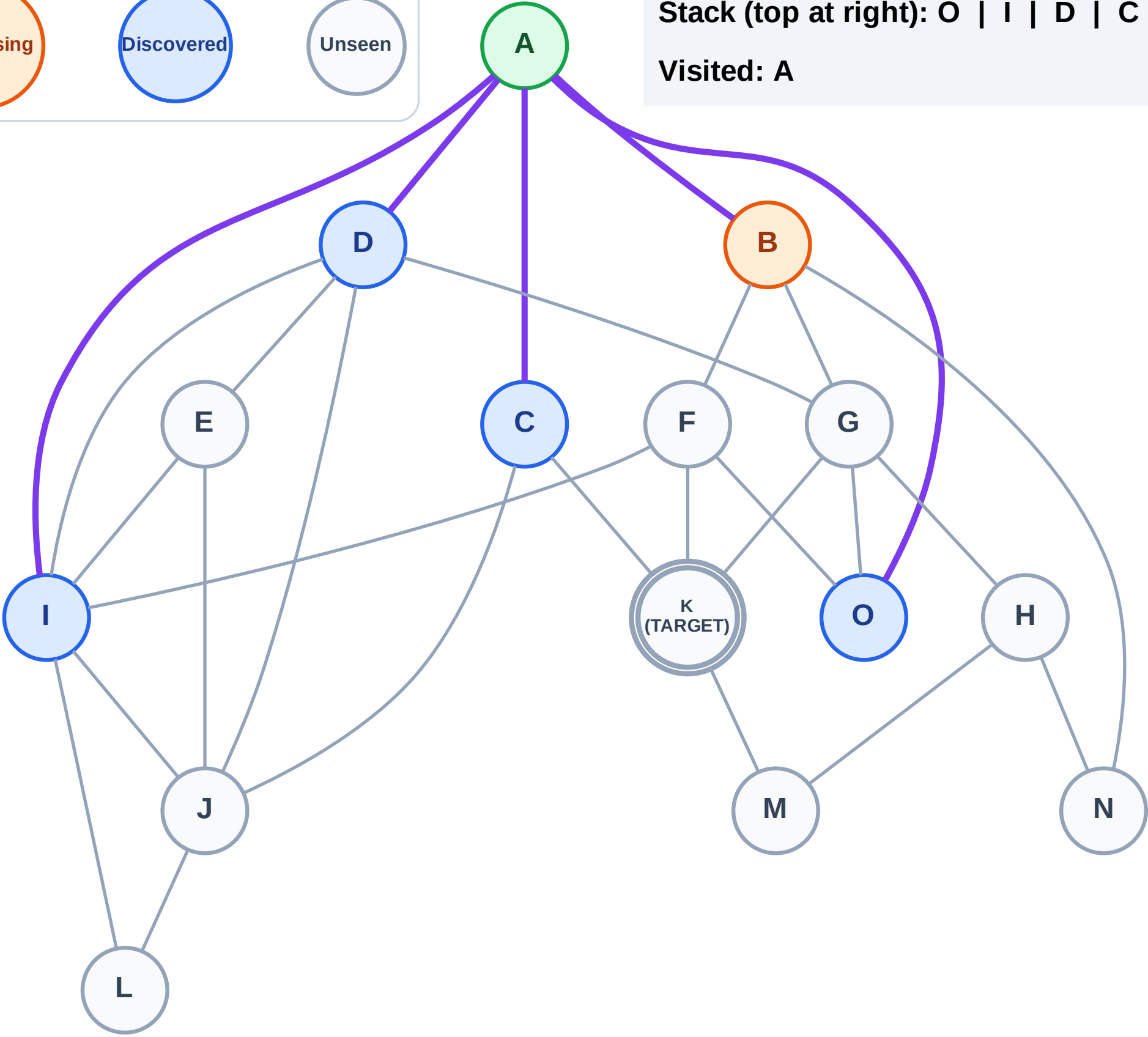
Processing

Discovered

Unseen

Stack (top at right): O | I | D | C

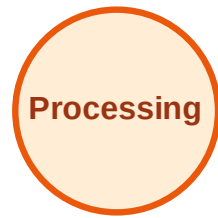
Visited: A



DFS Traversal

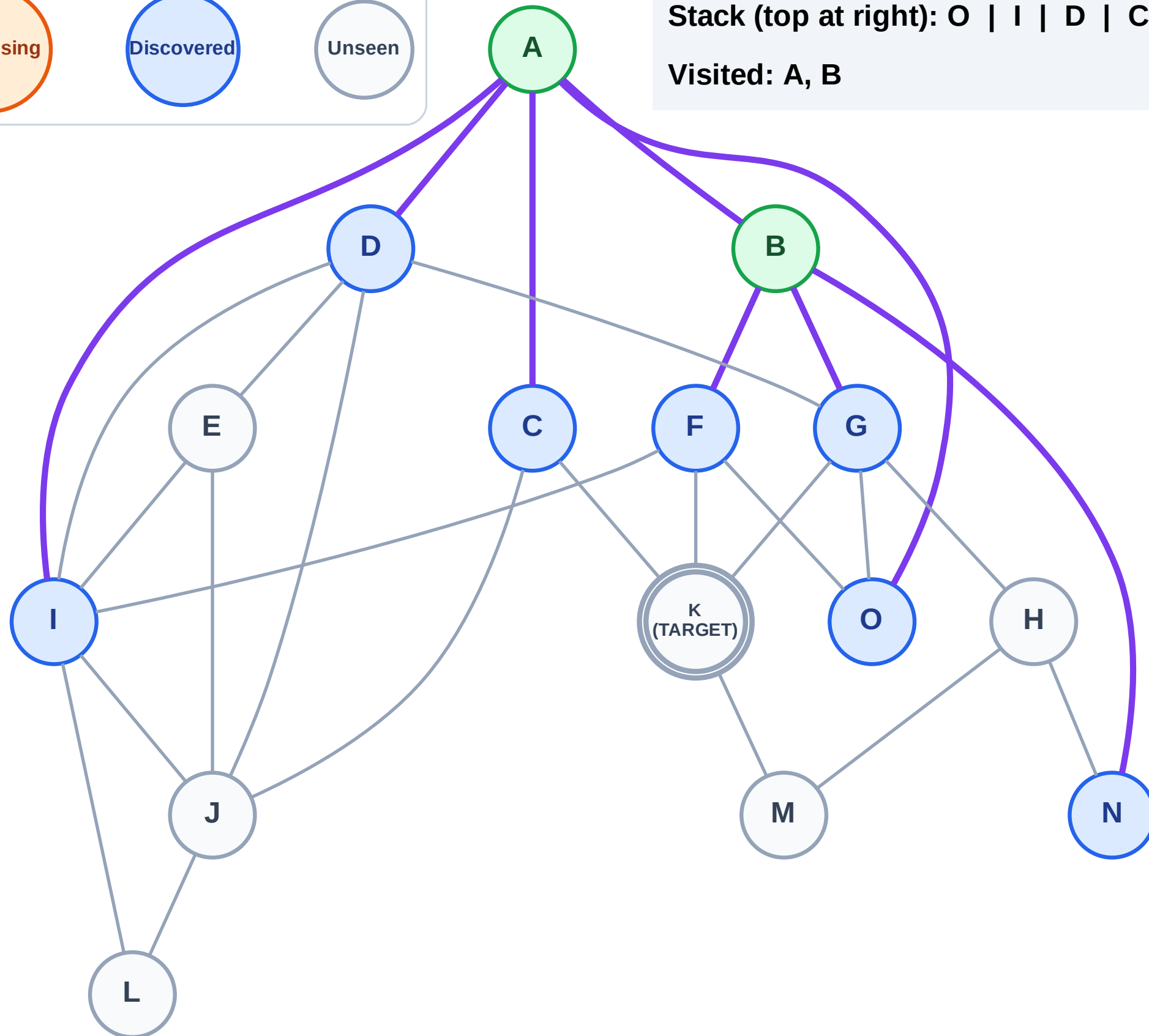
Step 5: Discover N, G, F from B

Legend



Stack (top at right): O | I | D | C | N | G | F

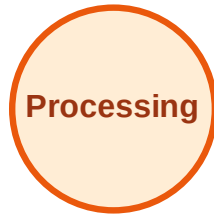
Visited: A, B



DFS Traversal

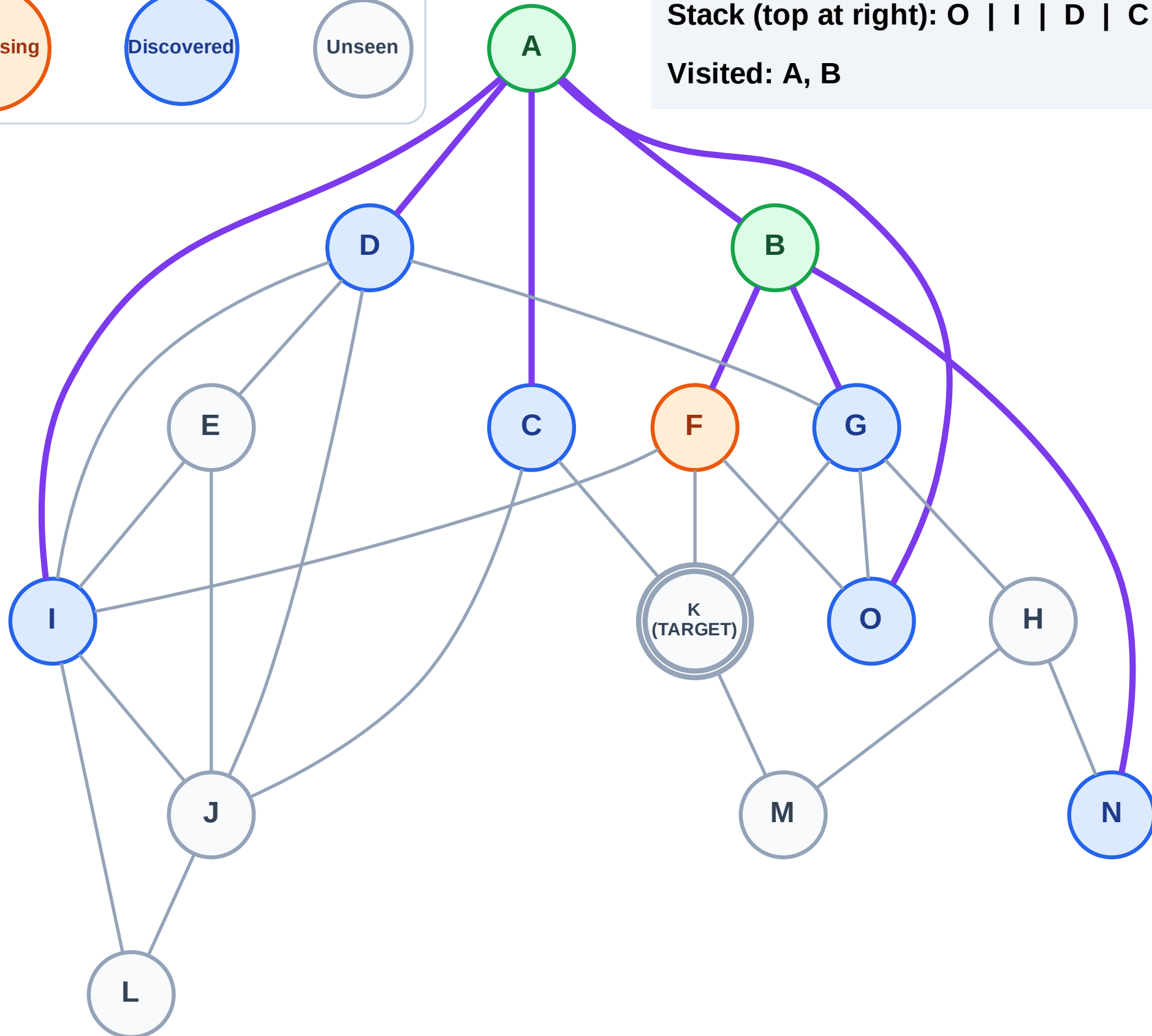
Step 6: Process node F

Legend



Stack (top at right): O | I | D | C | N | G

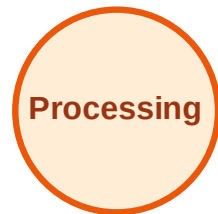
Visited: A, B



DFS Traversal

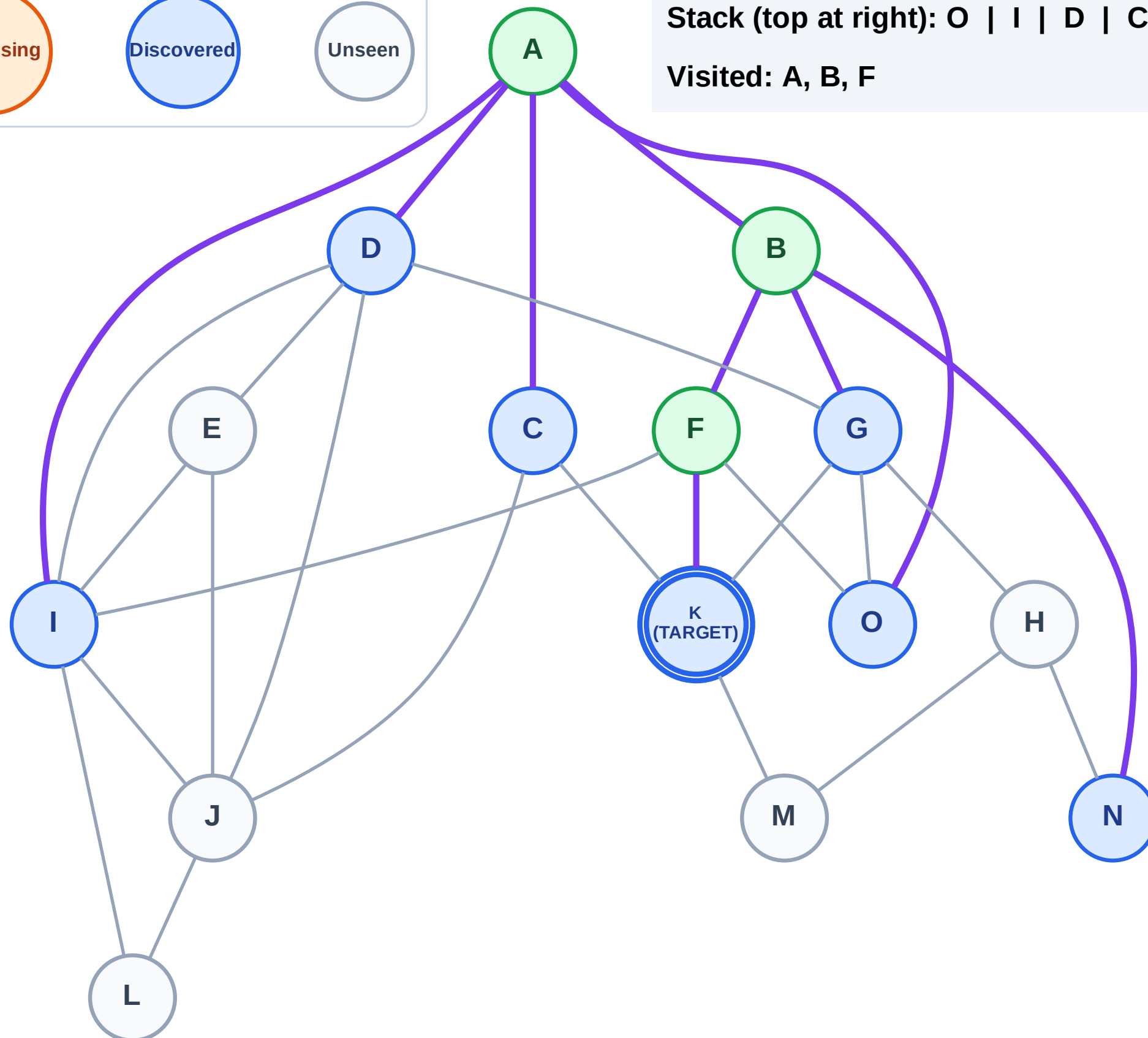
Step 7: Discover K from F

Legend



Stack (top at right): O | I | D | C | N | G | K

Visited: A, B, F



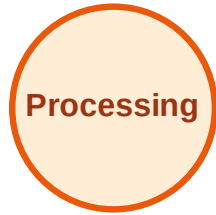
DFS Traversal

Step 8: Process node K

Legend



Complete



Processing



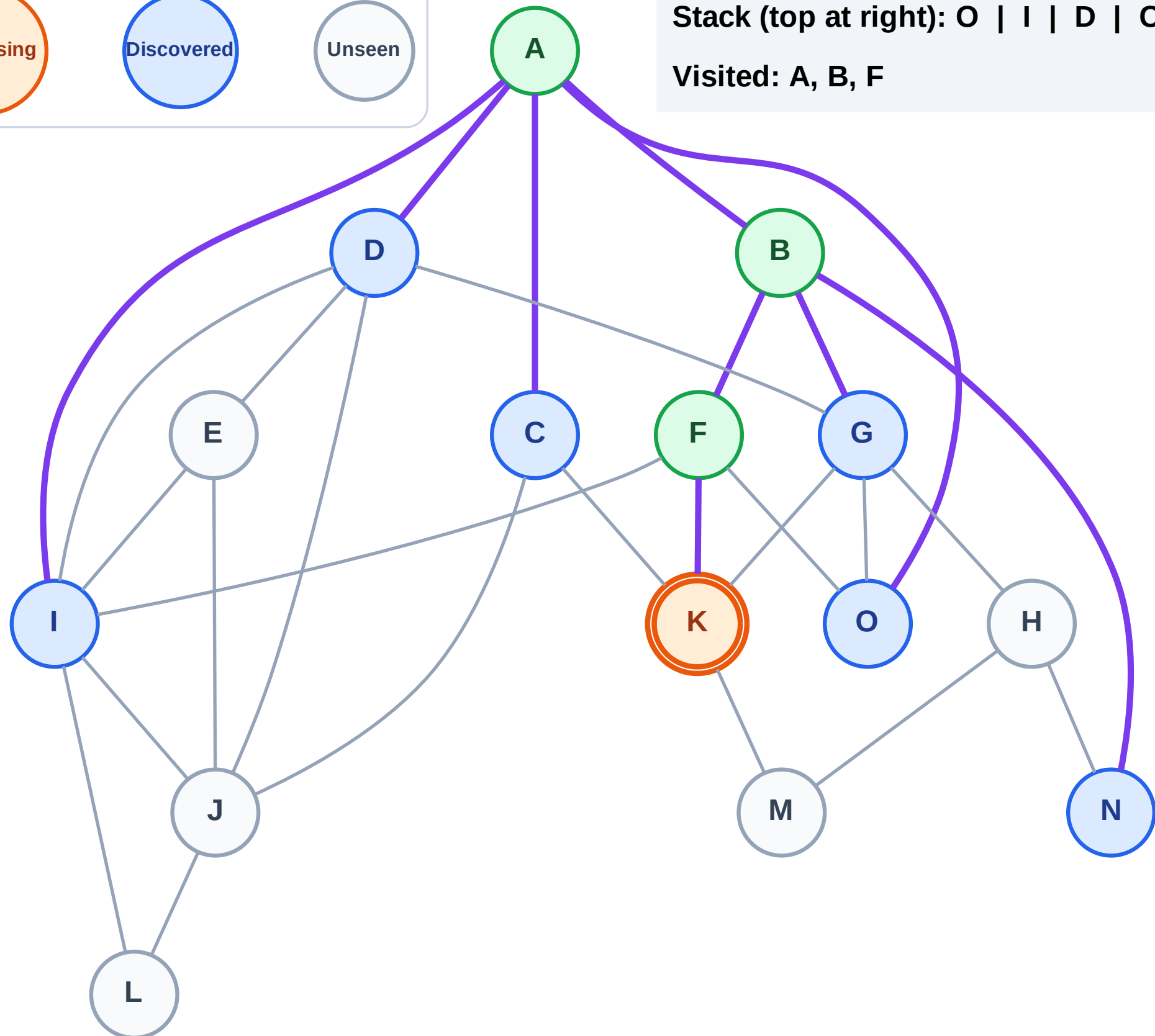
Discovered



Unseen

Stack (top at right): O | I | D | C | N | G

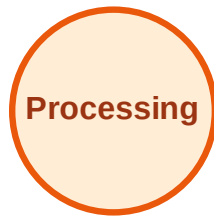
Visited: A, B, F



DFS Traversal

Step 9: TARGET NODE K REACHED! Path (3 hops): A → B → F → K

Legend



Stack (top at right): O | I | D | C | N | G

Visited: A, B, F, K

